Dynamic Sticky Real Relative Prices

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Abstract

I add a single friction to the real business cycle model and, without search or nominal rigidity, reproduce numerous empirical facts at once, including involuntary unemployment, a countercyclical labor share, a bust-heavy cycle with thick tails and faster recoveries, procyclical left-skewed utilization, and an acyclical real wage. The friction is a sticky, Hicksian adjustment of real relative prices, the rental and wage rates each moving with its own excess demand at a finite speed. I call the prices real because they chase their fundamental values and no inflation is involved. The two speeds of adjustment are set from wage durations and a plant survey of binding inputs. A firm with a thin profit share demands factors far more elastically than the household supplies them, so supply caps a boom and demand deepens a bust. The paper thus proposes an alternative approach to sticky prices that is monetarily neutral.

JEL: E32, E24, D50, E13, D24. **Keywords:** real business cycle; sticky real prices; short-side rationing; involuntary unemployment; business cycle asymmetry; labor share; capacity utilization.

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1 Introduction

This paper studies a real business cycle economy in which the two factor prices, the rental rate of capital and the wage, adjust toward their market-clearing values at finite, market-specific speeds. Trade takes place at the prices posted each quarter, and when a posted price differs from its clearing value the short side sets the quantity traded, so the long side is rationed. Everything else is the standard neoclassical growth model, with an optimizing household, competitive firms, decreasing returns to scale, and technology shocks scaled in the usual way.

A market that does not clear at once adjusts at some speed, and there is no reason for every market to adjust at the same speed. The classical account of a competitive market already has this form, with the price rising on excess demand and falling on excess supply at a finite rate (Samuelson, 1941), and with trade proceeding at prices that have not yet cleared (Hahn and Negishi, 1962). Instantaneous clearing is the special case in which an auctioneer completes this adjustment for all markets before any trade occurs, a device that is itself without foundation (Fisher, 1983; Herings et al., 1998). The economy here relaxes only the speed. Each real factor price rises while its market is short and falls while it is in surplus, at a finite speed of its own, with the shortage or surplus measured as in Hicks (1939).

Because a shock arrives every quarter, both posted prices always sit away from their clearing values, and trade takes place at them. The firm chooses capital and labor to maximize profit under decreasing returns, subject to the capital in place and the hours households offer, so where it would use more of a factor than exists it is held at that ceiling and the factor is fully employed, and where it would use less the suppliers hold the excess. That is the short-side rationing of the fixed-price disequilibrium tradition (Drèze, 1975; Benassy, 1975), given here an explicit law of motion for the price and applied to two factor markets.

Firm-side profit is very sensitive to factor prices and to productivity, in contrast to household utility, which is not, so the firm's factor demands are far more elastic than the household's supply of hours and saving. Where markets clear instantaneously, the household acts as a planner, dictating small and symmetric changes in production in response to productivity shocks. Under split minds and sticky prices, a favorable shock has the firm demanding far more than the household supplies, the supply binds, and output rises about as much as the household prefers. An adverse shock has the household supplying a little less but the firm demanding much less, both factors in surplus, and output falling far more than the household prefers and far more than it rises under a favorable shock. Booms are capped and busts deep, and foreseeing idle capital that earns no rental, the household invests less.

The data support this story on both the primitives and the results. The labor force and the capital stock, both household-side decisions, move a fraction as much as employment and capital utilization, both firm-side decisions once market clearing is gone, which explains the bust asymmetry results later.

The mechanism reaches four features of postwar US data at once. Unemployment is volatile and strongly countercyclical, while a competitive real business cycle model has none. The labor share is countercyclical, though under Cobb–Douglas technology and competitive pricing it is a constant, and here it moves because hired hours are paid the posted wage rather than their marginal product. The cycle is asymmetric, its busts deeper than its booms, whereas the frictionless economy is symmetric. Capacity utilization is procyclical and left-skewed.

The literature reaches each of these with a separate ingredient, a slow-moving wage in a search technology (Shimer, 2005), a cyclical redistributive shock to the labor share (Ríos-Rull and Santaeulàlia-Llopis, 2010), a binding capacity ceiling or a downward wage rigidity (Hansen and Prescott, 2005; Dupraz et al., 2025), a variable-utilization margin (Greenwood et al., 1988), and more in Section 2. Two recent papers instead let optimizing agents trade in markets that do not clear (Werning and Lorenzoni, 2026; Flynn et al., 2026). Neither is taken to the cycle's moments, which is what this paper asks of its one friction.

A reader places this posted price beside the Calvo and Rotemberg pricing of the New Keynesian model (Calvo, 1983; Rotemberg, 1982), but the economics differs on four counts. The sticky object here chases a real target that moves with technology, not a nominal one that moves with expected inflation, so the model has no inflation. Their reset price is forward looking. This rule closes a fixed fraction of the current gap and looks back, and rational expectations over the future distortions, which chase the technology shock, enter only the household's saving decision. They distort the output price relative to marginal cost, a markup. This rule opens a factor-price wedge they lack. Their primitive is menu costs and market power, mine the relative speed at which each market answers its own surplus or shortage.

Two market-clearing speeds are the model's novel disciplined parameters. The wage's speed is assigned from external micro evidence that wages stay fixed for about six quarters (Grigsby et al., 2021), a wage half-life of 0.95 year. The rental's speed is fitted to a single plant-level moment, the ratio of labor-constrained to capital-constrained plants in the US Census Quarterly Survey of Plant Capacity Utilization (QPC), a ratio of 1.55 in the model against 1.54 in the data. The calibration then finds that the capital market clears about three times as fast as the labor market, a rental half-life of 0.31 year.

The friction reproduces the four features on moments it never targets, from posted prices

a third of a percent from their competitive values on average. Cyclical unemployment has a standard deviation of 1.48 percentage points against 0.95 in the data and a correlation with output of −0.98 against −0.85, a within-period rationing rate against the data's labor-force rate, with no search, matching, or bargaining technology. The labor share correlates with output at −0.44 against −0.26. Output has a skewness of −1.10 against −1.05 and thick tails, busts about twice as volatile as booms, from symmetric shocks and no asymmetric adjustment cost. Capacity utilization has a skewness of −1.30 against −0.80 and correlates with output at 0.95 against 0.89.

The reach extends beyond the four, none of it targeted. On the cycle's shape, busts fade faster than booms as in the plucking evidence, and output, investment, and unemployment have the thick tails of the data. On prices, the real wage is nearly acyclical, the Dunlop–Tarshis fact, and the labor share is as persistent as in the data. On quantities, the supplies move a fraction as much as the factors employed, as the labor force and the capital stock do beside employment and utilization, and the persistence of the investment rate, the relative volatility of consumption, and the skewness and persistence of unemployment all move toward the data. Two of these, the recovery asymmetry and the labor-share correlogram, carry literatures of their own. The friction earns this reach at a cost, leaving output's persistence too low and the volatility of investment and of the wage too small.

Section 2 places the paper in the literature, Section 3 states the model with the full optimization problem for every agent, Section 4 maps it to the data, Section 5 characterizes the mechanism on the calibrated model, Section 6 sets its implications against the data and varies the assigned parameters, and Section 7 concludes.[1]

2 Related literature

This paper's device is short-side rationing at a non-clearing price, the core of the fixed-price disequilibrium tradition (Drèze, 1975; Benassy, 1975). Drèze reaches equilibrium "by introducing quantity constraints on the net trades of those commodities for which the price constraints are binding," and Benassy (1975) feeds "quantity signals into demand functions, in addition to prices." Barro and Grossman (1971) put it in the goods and labor markets at once, and Barro (2025) restates the case, where a real wage away from clearing in either direction lowers output. Malinvaud (1977) is its founding statement for unemployment, sorting the Keynesian, classical, and repressed-inflation regimes by which market is short, with each price fixed within the period and moving autonomously between periods, and its modern general-equilibrium treatment reaches a Walrasian equilibrium only in the long run

[1] Appendices A and B follow the text. Appendices OA.1 to OA.5 are in the online supplement.

(Herings et al., 1998). I keep the rationing and replace the autonomous price with one that moves at a finite speed toward its market-clearing value, closing a fixed fraction of the gap each quarter, and apply it to both factor markets at once.

Werning and Lorenzoni (2026) give finite-speed adjustment a forward-looking general-equilibrium foundation, a price moving in proportion to a discounted average of its own excess demand with the frictionless allocation as the rest point. Schoder (2020) is a quantitative model in the same tradition, where a non-clearing nominal wage leaves unemployment as demand-determined job rationing on the labor market alone. Flynn et al. (2026) instead have the firm commit to a price and its inputs before demand is known, so its product market is slack or rationed ex post and clears again as uncertainty vanishes. Here the rationing is dynamic, dies as the price speeds up, reaches two factor markets rather than one, and needs no uncertainty. Estimated sticky-wage models drop the same short side, so households are made to work more than they offer and firms to sell below marginal cost (Huo and Ríos-Rull, 2020; Holden, 2025).

The involuntary unemployment that results is job rationing in the sense of Michaillat (2012), the unemployment that survives when matching frictions are switched off, obtained there from a rigid real wage. I reach the same object from a real wage that adjusts at finite speed, alongside a mirror-image margin on capital. That capital margin is the product-market twin of Michaillat and Saez (2015), whose matching friction leaves firms selling below capacity, except that mine is a two-sided factor-price wedge, holding a factor below its marginal product in booms and above it in busts. Its one-sided version is the occasionally binding capacity ceiling of Hansen and Prescott (2005), which caps booms alone where my convex kink binds in both directions.

Unemployment is volatile and strongly countercyclical, which a competitive real business cycle model cannot produce, and the dominant account is search and matching (Mortensen and Pissarides, 1994). Shimer (2005) showed that its textbook calibration understates the volatility of unemployment and vacancies, and Hall (2005) restores it with a real wage held sticky inside the bargaining set, as does raising the worker's value of nonmarket time (Hagedorn and Manovskii, 2008). I supply that rigidity as a primitive, and a Greenwood–Hercowitz–Huffman (GHH) outside option makes the flow opportunity cost of work procyclical and more volatile than the marginal product (Chodorow-Reich and Karabarbounis, 2016).

The labor share is countercyclical, a constant under Cobb–Douglas technology and competitive pricing, and the literature moves it with a cyclical redistributive shock in an otherwise competitive model (Ríos-Rull and Santaeulàlia-Llopis, 2010), a falling relative price of investment (Karabarbounis and Neiman, 2014), or cyclical markups (Cantore et al., 2021).

Choi and Ríos-Rull (2009) move it with a noncompetitive wage set by bilateral Nash bargaining. Mine falls out of a rationed wage paid below the marginal product, with no shock to shares, no change in technology and no bargaining, the wage being noncompetitive only in the speed at which it closes its gap.

The same wedge sets the paper beside the misallocation literature (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009), whose object is a static cross-sectional spread of marginal products, and beside the business-cycle-accounting labor wedge (Chari et al., 2007; Karabarbounis, 2014). Asker et al. (2014) shows dynamic adjustment frictions can generate measured misallocation with no true distortion. Mine is an intertemporal, economy-wide relative price common to all firms, and unlike a monopsony markdown (Berger et al., 2022) it is a price-taking gap, transitory and two-sided.

The cycle is asymmetric, its busts deeper than its booms, and the closest published account is the plucking model of Dupraz et al. (2025), where contraction amplitude forecasts the following expansion but not conversely, obtained there from downward nominal wage rigidity in a search economy. I obtain the same dynamics from a purely real friction, the convex kink in the rationing loss, so the asymmetry needs neither a nominal rigidity nor a concave hiring rule (Ilut et al., 2018). Its statistics follow the deepness and steepness decomposition of Neftçi (1984) and Sichel (1993). Thick tails are the other mark of nonlinearity, and Khan and Thomas (2008) make the skewness and excess kurtosis of the aggregate investment rate the test of lumpy investment, finding both washed out by market-clearing prices and near zero in the annual data. Here the tails come from the rationing loss, and the model's excess kurtosis lies beside that of the quarterly cyclical series.

Capacity utilization is procyclical and left-skewed, which the literature obtains from a variable-utilization margin (Greenwood et al., 1988) or from unobserved input variation behind procyclical productivity (Basu and Kimball, 1997). Mine reaches idle capital from a single relative-price wedge rather than a utilization choice, and its defining claim, that one factor or the other is occasionally rationed, meets firms' own reports of which input limits production. The US Census Quarterly Survey of Plant Capacity Utilization asks exactly this, and the ratio of labor-limited to equipment-limited plants identifies the rental's adjustment speed. In the same survey Gorodnichenko and Shapiro (2011) find capital idle most of the week. Where the literature reaches each of these four regularities by a separate addition to the standard model, the single friction here reaches all four together, and every one vanishes in the frictionless limit.

Because the friction is a sticky price, a reader reaches for the Calvo and Rotemberg pricing of the New Keynesian model (Calvo, 1983; Rotemberg, 1982), and the four differences of Section 1 are economic rather than technical. Where their friction sits on a single price,

the output price over marginal cost, mine sits on the relative price of two factors, and where their question is how market power and menu costs make a nominal price sticky, mine is how fast each factor market absorbs its own surplus or shortage. A real rigidity in Ball and Romer (1990) works only alongside a nominal friction, and this one needs none.

A parallel New Keynesian literature makes unemployment a labor-market friction, off a wage Phillips curve (Galí, 2011) or from search effort "that does not require monopoly power" (Christiano et al., 2021). The closest structural relative, Erceg et al. (2000), staggers nominal contracts whose households hold wage-setting power and leave their labor-supply schedule by choice, where mine leaves it involuntarily. The objection of Barro (1977), that parties can contract around a sticky wage, does not bite on an economy-wide relative price faced by every agent as a price-taker.

The quantitative discipline is that of the real business cycle program (Kydland and Prescott, 1982; Cooley and Prescott, 1995), whose convention I adopt, restricting the technology shock so the evaluated model matches observed output volatility and reporting every moment from the full calibrated nonlinear model. Decreasing returns, the model's second ingredient, is the source of the strictly positive pure-profit share and lets it reproduce that share and data-scale unemployment volatility together, the same way the search literature reconciles a large average surplus with a small marginal one, as the decreasing-returns technologies of Khan and Thomas (2008) and Khan and Thomas (2013) do. What this paper adds to the class is a single real friction that turns the frictionless economy into a model with involuntary unemployment while keeping the class's discipline intact.

3 Model

Time is discrete, one period is a quarter, and the economy is closed, with one good, a representative household and a representative firm. Output is the numeraire, so the only prices are the two real factor prices, the rental rate of capital services r and the real wage w, both in units of output, and no nominal price is set anywhere. The household owns the capital stock and the firm, renting capital to the firm and receiving its pure profit lump sum, and there are no other assets.

Following Cooley and Prescott (1995), letter case separates a household-controlled quantity from its firm counterpart. Lower-case k is the capital the household owns and accumulates, upper-case K the capital the firm employs, with $\varkappa \equiv K/k \in (0, 1]$ the rented fraction. Labor is marked by a superscript, N^s the household's notional offer and N the hours the firm employs, and the firm takes only the rationed short side, $N \leq N^s$ and $K \leq k$. The capital shortfall $k - K > 0$ is idle capital, with idle fraction $1 - \varkappa$, and the labor shortfall $N^s - N > 0$

is involuntary unemployment, with rate $u = \max(N^s - N, 0)/N^s$, so $N/N^s = 1 - u$ is the employment rate.

Total factor productivity is the model's only exogenous driver, and it follows a quarterly AR(1) in logs,

$$\log z_t = \rho_z \log z_{t-1} + \sigma_z \varepsilon_t, \qquad \varepsilon_t \sim N(0,1) \text{ iid}, \tag{1}$$

with persistence ρ_z and per-quarter innovation standard deviation σ_z, both set in Section 4. The shock is aggregate, with no idiosyncratic shocks, so households are ex-ante identical and the representative-household reduction is exact.

3.1 Households

The household is the model's only intertemporal optimizer, choosing consumption c_t, next-period capital k_{t+1}, and labor supply N_t^s to maximize

$$\max_{\{c_t,\, k_{t+1},\, N_t^s\}_{t\geq 0}} \; \mathbb{E}_0 \sum_{t\geq 0} (1+\rho)^{-t} \log\Big(c_t - \chi \, \frac{(N_t)^{1+\nu}}{1+\nu}\Big), \tag{2}$$

whose log argument is the GHH composite $\tilde{c}_t \equiv c_t - \chi (N_t)^{1+\nu}/(1+\nu)$, consumption net of the disutility of hired hours (Greenwood et al., 1988; Jaimovich and Rebelo, 2009; Ríos-Rull and Santaeulàlia-Llopis, 2010). Here ρ is the rate of time preference, δ the quarterly depreciation rate, χ the labor-disutility scale and ν the inverse Frisch elasticity of labor supply.

The household faces the flow budget constraint, the GHH admissibility condition, and non-negativity,

$$\begin{aligned} k_{t+1} &= (1-\delta)\, k_t + r_t\, K_t(z_t, k_t, r_t, w_t) + w_t\, N_t(z_t, k_t, r_t, w_t, N_t^s) + \pi_t(z_t, k_t, r_t, w_t) - c_t, \\ c_t &\geq \chi \, \frac{(N_t)^{1+\nu}}{1+\nu}, \qquad k_{t+1} \geq 0. \end{aligned} \tag{3}$$

The realized demands K_t and N_t in the budget are the firm's rationed demands from (8), fixed by the short-side rule the household faces,

$$K_t(z_t, k_t, r_t, w_t) \leq k_t, \qquad N_t(z_t, k_t, r_t, w_t, N_t^s) \leq N_t^s. \tag{4}$$

This is a fixed-price equilibrium in the sense of Drèze (1975), so the resulting unemployment $N_t^s - N_t \geq 0$ is involuntary rationing at a posted wage. The optimal path satisfies the transversality condition $\lim_{T\to\infty} \mathbb{E}_t[\beta^T k_{T+1}/\tilde{c}_T] = 0$.

The Greenwood et al. (1988) specification removes the wealth effect on hours,[2] so the labor margin depends on the wage alone and the saving and employment decisions separate. The household posts a notional offer at the wealth-free margin,

$$\chi(N^s)^\nu = w \quad \Longrightarrow \quad N^s = (w/\chi)^{1/\nu}. \tag{5}$$

The offer enters only as the ceiling $N_t \le N_t^s$. Posting $N_t^s = (w_t/\chi)^{1/\nu}$ is weakly dominant state by state, since the hours term $g(N) = w_t N - \chi N^{1+\nu}/(1+\nu)$ in the composite peaks at that margin and the atomistic household does not internalize the effect of its offer on the firm's demand for its capital.[3]

The household's intertemporal condition is the Euler equation,

$$\tilde{u}'(\tilde{c}_t) = (1+\rho)^{-1}\,\mathbb{E}_t\big[\big(1-\delta+r_{t+1}\,\varkappa_{t+1}\big)\,\tilde{u}'(\tilde{c}_{t+1})\big], \qquad \varkappa_t \equiv \frac{K_t}{k_t} \in (0,1], \tag{6}$$

on the GHH composite $\tilde{c}_t$, with $\tilde{u}' = 1/\tilde{c}$ under log utility. The rented fraction $\varkappa_{t+1}$ carries the friction in, since the household collects the rental only on the fraction of its capital the firm rents, so it prices saving against the discounted rental $r_{t+1}\varkappa_{t+1}$, not the posted rental r_{t+1}, and, being atomistic, it takes $\varkappa_{t+1}$ as given.

3.2 The firm

The firm is the agent the friction acts on. Technology is Cobb–Douglas with decreasing returns to scale,

$$Y = z\,K^\alpha N^\beta, \qquad \gamma = \alpha+\beta < 1, \tag{7}$$

where α and β are the capital and labor exponents and the residual $1-\gamma$ is the income share of a fixed factor, which shows up as positive pure profit. The firm hires K and N to maximize revenue net of factor payments, taking the posted prices (r, w) and the available quantities as given,

$$\max_{K,\,N} \;\; z\,K^\alpha N^\beta - rK - wN \qquad \text{subject to} \qquad 0 \le K \le k, \quad 0 \le N \le N^s(w). \tag{8}$$

[2] Were a wealth effect on labor supply present, hours offered would fall as wealth rises, so the offer would move with wealth over the cycle and the ergodic frequencies of the labor-rationing regimes would change. That comparison lies outside the present GHH specification.

[3] The firm's capital demand does rise with the hours it can hire, so a household that internalized the link would offer more hours to have more of its capital employed. Being one of many suppliers, it takes the firm's demands as given.

It is a box-constrained profit maximizer that takes r and w as given and is rationed by the ceilings $K \le k$ and $N \le N^s(w)$. It observes z, k, and (r, w), hence the labor ceiling $N^s = (w/\chi)^{1/\nu}$. The traded quantity of each factor is the firm's choice, so the short side of a market is the firm's demand where its ceiling is slack and the supply where it binds.

Strict concavity ($\gamma < 1$) on the compact box gives a unique maximizer, characterized by four Karush–Kuhn–Tucker (KKT) regimes named by their surplus content, a factor in surplus where the firm wants less than exists and in shortage where it wants more,

$$\begin{aligned} &\text{capital in shortage} \iff K = k, \quad \text{capital in surplus} \iff K < k \ \ (k - K \text{ idle});\\ &\text{labor in shortage} \iff N = N^s, \quad \text{labor in surplus} \iff N < N^s \ \ (N^s - N \text{ unemployed}). \end{aligned} \tag{9}$$

The four regimes are

$$\begin{aligned} \text{reg}_\emptyset &: \ K < k \ \text{ and } \ N < N^s, \ \ \text{(both factors in surplus, interior)},\\ \text{reg}_K &: \ K = k \ \text{ and } \ N < N^s, \ \ \text{(capital in shortage, labor in surplus, unemployment)},\\ \text{reg}_N &: \ K < k \ \text{ and } \ N = N^s, \ \ \text{(labor in shortage, capital in surplus, idle capital)},\\ \text{reg}_{KN} &: \ K = k \ \text{ and } \ N = N^s, \ \ \text{(both factors in shortage)}. \end{aligned} \tag{10}$$

A factor in surplus satisfies its first-order condition and a factor in shortage sits at its ceiling,

$$\alpha z K^{\alpha-1} N^{\beta} = r \quad \text{if } K < k, \qquad \beta z K^{\alpha} N^{\beta-1} = w \quad \text{if } N < N^s, \tag{11}$$

so in $\text{reg}_\emptyset$ both conditions hold and give the unconstrained demands K^d and N^d, the factors the firm would employ if neither ceiling bound, and in reg_N the capital condition holds at $N = N^s$ and gives the conditional capital demand $K^\dagger$, the capital the firm wants when its hours are capped at the offer, below K^d because it cannot get the hours to work more capital,

$$\begin{aligned} K^d(z, r, w) &= \left[z\, \alpha^{1-\beta} \beta^{\beta}\, r^{-(1-\beta)} w^{-\beta} \right]^{\frac{1}{1-\gamma}}, \qquad N^d(z, r, w) = \left[z\, \alpha^{\alpha} \beta^{1-\alpha}\, r^{-\alpha} w^{-(1-\alpha)} \right]^{\frac{1}{1-\gamma}},\\ K^\dagger(z, r, N^s) &= \left[\alpha z (N^s)^{\beta} / r \right]^{\frac{1}{1-\alpha}}, \end{aligned} \tag{12}$$

in reg_K the labor condition holds at $K = k$, and at the corner neither holds. The firm evaluates all four candidates and takes the highest-profit feasible one, so the realized pair is the joint profit-maximizing allocation, not the market-by-market smaller of unconstrained demand and ceiling as in Drèze (1975) and Benassy (1975). Appendix OA.4 gives the candidate set, and Section 5 reports the average size and cyclical asymmetry of the rationing.

3.3 Sticky real factor prices

The posted prices chase the market-clearing values at the current state, defined first. A superscript CE denotes the value an object would take if both factor prices cleared within the period at the current (z, k), the predetermined stock k included. It is a static object, recomputed every quarter at whatever state the economy is in, and distinct from the frictionless economy of Subsection 3.5, in which prices clear every quarter and the stock itself follows a different path. The clearing rental $r^{\mathrm{CE}}(z,k)$, wage $w^{\mathrm{CE}}(z,k)$ and employment $N^{\mathrm{CE}}(z,k)$ solve

$$\underbrace{\alpha z k^{\alpha-1}(N^{\mathrm{CE}})^{\beta} = r^{\mathrm{CE}}}_{\text{capital demand } =k}, \qquad \underbrace{\beta z k^{\alpha}(N^{\mathrm{CE}})^{\beta-1} = w^{\mathrm{CE}}}_{\text{firm's labor demand}},$$
$$\underbrace{\chi(N^{\mathrm{CE}})^{\nu} = w^{\mathrm{CE}}}_{\text{household's labor supply}}, \tag{13}$$

three equations in the three unknowns $(r^{\mathrm{CE}}, w^{\mathrm{CE}}, N^{\mathrm{CE}})$ at the state (z, k). They are analytic at every state, and their cyclical loadings, the elasticities of r^{CE} and w^{CE} with respect to z, follow in Appendix OA.1.

Away from these values each market carries a surplus or a shortage, measured after the other market has cleared. Given a rental r, let $\check{w}(z,r)$ be the wage at which hours would clear, $N^d(z,r,\check{w}) = N^s(\check{w})$, and given a wage w, let $\check{r}(z,k,w)$ be the rental at which the firm would rent exactly the stock, $K^d(z,\check{r},w) = k$. Each is the value the other price would need to clear its own market, and it equals the clearing value only when the given price is at clearing, $\check{w}(z, r^{\mathrm{CE}}) = w^{\mathrm{CE}}$ and $\check{r}(z,k,w^{\mathrm{CE}}) = r^{\mathrm{CE}}$. The two log excess demands are then

$$\mathcal{E}_r \equiv \log K^d\big(z, r, \check{w}(z,r)\big) - \log k, \qquad \mathcal{E}_w \equiv \log N^d\big(z, \check{r}(z,k,w), w\big) - \log N^s(w), \tag{14}$$

so that $\mathcal{E}_j > 0$ is a shortage and $\mathcal{E}_j < 0$ a surplus, and $\mathcal{E}_w$ is the short-run excess demand for hours with the capital stock fully used. This is the measure Hicks (1939) takes as the relevant one for the stability of a market, excess demand "when other prices are adjusted so as to preserve equilibrium in the other markets."[4]

The model's one friction is that factor prices adjust slowly to these imbalances. Within each quarter each log price moves in proportion to its own market's excess demand at that

[4]Hicks (1939, p. 66) asks whether a fall in a price must make demand exceed supply when the other prices are given, or when they are adjusted to keep their own markets in equilibrium, and holds that the second is what matters for stability. The first alternative of Appendix A.3 measures excess demand in the first way, at the posted prices.

quarter's state (z_t, k_t), with $\tau \in [0,1]$ the time within the quarter,

$$\frac{\mathrm{d}\log p_j(\tau)}{\mathrm{d}\tau} = \omega_j \,\mathcal{E}_j\big(\log p_j(\tau); z_t, k_t\big), \qquad \log p_j(0) = \log p_{j,t-1}, \qquad \log p_{j,t} = \log p_j(1), \quad (15)$$

for $j \in \{r, w\}$, the adjustment of a single market in Samuelson (1941), here at a finite flexibility $\omega_j > 0$, the rate at which the log price moves per log point of imbalance. The adjustment is a mechanical law of motion, and no agent chooses it.

Two implications follow, both exact. First, because the demands of (12) are log-linear in the two prices, the stock is predetermined and the labor offer is isoelastic, each excess demand is linear in its own log price gap,

$$\mathcal{E}_j = -\psi_j\big(\log p_j - \log p_j^{\mathrm{CE}}(z,k)\big), \qquad \psi_r = \frac{1+(1-\beta)/\nu}{1-\alpha+(1-\gamma)/\nu}, \qquad \psi_w = \frac{1}{1-\beta}+\frac{1}{\nu}. \quad (16)$$

The imbalance elasticity ψ_j is the log excess supply that a price one percent above clearing creates, for labor the firm hiring less along its short-run demand and the household offering more. Second, the clearing values stay fixed within the quarter, since (z_t, k_t) does and the labor offer carries no wealth effect, so (15) integrates exactly (Appendix A.1). Each factor price therefore closes a fixed share of the distance to its clearing value every quarter,

$$\log r_t = \phi_r \log r_{t-1} + (1-\phi_r)\log r^{\mathrm{CE}}(z_t,k_t), \qquad \log w_t = \phi_w \log w_{t-1} + (1-\phi_w)\log w^{\mathrm{CE}}(z_t,k_t), \quad (17)$$

with persistence $\phi_j = e^{-\omega_j\psi_j} \in (0,1)$, and its imbalance closes at the same rate.

The persistence ϕ_j is the share of a price gap still open after a quarter and $1-\phi_j$ the share closed, so the half-life $\ln(0.5)/\ln\phi_j$ is the time it takes half of a gap to close, and $-\ln\phi_j = \omega_j\psi_j$ is the speed, the price's flexibility times the imbalance elasticity. Within the quarter a gap shrinks in proportion to its size, so it never closes in finite time, and a price clears its market within the quarter only in the limit $\omega_j \to \infty$, where $\phi_j \to 0$. Each gap is a stationary AR(1) with root ϕ_j. The two markets carry separate persistences, which Section 4 pins to the data.

The same rule results from an intermediary who weighs posting away from the clearing price against the cost of changing its post (Appendix A.2). Appendix A.3 turns to two other measures of the imbalance and calibrates each to the same targets. One takes each market's excess demand at the prices actually posted in both markets, without letting the other market clear first. The other takes the firm's effective demand, its demand in one market given the constraint it meets in the other. Most results hold under either, with unemployment volatile and countercyclical, the labor share countercyclical and the cycle

bust-heavy, while the real wage turns procyclical.

Define the two competitive price gaps $x_r = \log(r_t/r^{\mathrm{CE}}(z_t, k_t))$ and $x_w = \log(w_t/w^{\mathrm{CE}}(z_t, k_t))$, the log deviations of the posted prices from their contemporaneous competitive values. Because r^{CE} and w^{CE} are the marginal products, each gap measures a factor price against its own marginal product, and with the good as numeraire the two are independent relative prices, so both can be off their competitive values at once. They are zero under instantaneous clearing and are the two persistent states the household learns through (z_t, k_t, r_t, w_t).

3.4 Timing

Events run in order. The shock z_t is realized against the predetermined stock k_t. The two prices then adjust by (15) to (r_t, w_t). The household posts its notional offer N_t^s against the posted wage, the firm hires, and (N_t, Y_t) is realized, and the household then splits output into consumption c_t and next-period capital k_{t+1}.

3.5 Recursive equilibrium

The aggregate state each quarter is (k, z, r, w), the predetermined capital stock, current productivity, and the two prices posted at the start of the quarter.

A *recursive sticky-relative-price equilibrium* is a set of functions

$$\big(c,\ k',\ N^s,\ K,\ N,\ r',\ w',\ \varkappa,\ u\big)$$

of the state, the household policies $c(k, z, r, w)$ and $k'(k, z, r, w)$ and offer $N^s(w)$, the firm's demands $K(k, z, r, w)$ and $N(k, z, r, w)$, the price functions $r'(r, z', k')$ and $w'(w, z', k')$, the rented fraction $\varkappa = K/k$ and the unemployment rate $u = \max(N^s - N, 0)/N^s$, such that

(i) (c, k') solve the household's problem (2) subject to (3) and (4), given (1), the posted prices, the firm's demands (K, N), and the rented fraction $\varkappa'$ in the gross return $R' = 1 - \delta + r'\varkappa'$;

(ii) (K, N) solve the firm's problem (8) given (r, w) and the ceilings k and $N^s(w)$, and are the quantities traded;

(iii) r' and w' follow the adjustment (15) from (r, w) at the state (z', k');

(iv) the goods market clears, $c + k' - (1 - \delta)k = Y$;

(v) productivity follows (1), the posted prices follow (iii), and capital follows the household's policy k'.

No factor market is required to clear. The *frictionless economy* is the same economy with $\phi_r = \phi_w = 0$, so that both prices sit at their clearing values every quarter, no ceiling binds, $u = 0$ and $\varkappa = 1$, and the allocation is the competitive one.

The within-quarter allocation exists and is unique by strict concavity on the compact box (Appendix OA.4), and the household's recursive problem has a unique solution for a given aggregate law of motion (Appendix OA.4.1). Whether the equilibrium as a whole is unique is not established analytically, the map from an aggregate saving law into the household's best response not being a contraction in general. Numerically the iteration returns the same saving policy from unrelated starting points (Appendix OA.5). With $\rho_z, \phi_r, \phi_w \in (0, 1)$ the productivity process and the two price gaps are stationary, the time-iterated saving policy converges (Appendix OA.5), and the simulated capital stock settles to a stationary ergodic distribution on which every reported moment is computed (Appendix OA.5.3).

4 Mapping the model to the data

4.1 Externally assigned parameters

Standard parameters. The model is calibrated to US data, externally set parameters first and three more inside the model after. Six parameters are set outside the cyclical model, each to a conventional steady-state moment or a normalization in the manner of Cooley and Prescott (1995) and Kydland and Prescott (1982),[5] none to fit a cyclical moment, and listed with source in Table 1, the returns to scale $\gamma = 0.90$ fixing the pure-profit share $\pi/Y = 1 - \gamma = 0.10$.

Wage rigidity. The measured object is a base-wage duration. Using administrative payroll data that isolate the base contract wage, Grigsby et al. (2021) find that "the typical worker who remains continuously employed on the same job" has a base-wage duration of "about six quarters," and Barattieri et al. (2014) corroborate that order of magnitude on measurement-error-corrected Survey of Income and Program Participation (SIPP) data, with a within-job wage-change probability "between 16.3 and 21.6 percent per quarter," a constant-hazard duration of 4.6 to 6.1 quarters within jobs.

The duration-to-persistence conversion of Appendix B turns that spell length into the persistence of the rule (17), under the same constant-hazard mapping by which the duration

[5]The deterministic steady state sets $\sigma_z = 0$, so $z = 1$, the posted prices sit at their competitive values and the firm is unrationed. The Euler equation gives $r_{ss} = \rho + \delta$, the GHH margin $\chi(N_{ss})^{\nu} = w_{ss}$, and the accumulation identity $c_{ss} = Y_{ss} - \delta k_{ss}$. The friction operates purely through fluctuations, so ϕ_r, ϕ_w leave it unchanged.

Table 1: Externally set parameters.

description	parameter	value	source
returns to scale	γ	0.90	pure-profit share $1-\gamma = 0.10$, assigned. Basu et al. (2006) estimate returns to scale of 0.89 in nondurable manufacturing and Khan and Thomas (2008) set $\alpha + \nu = 0.896$. Sensitivity in Appendix OA.3
capital and labor exponents	$\alpha,\ \beta$	0.324, 0.576	factor split 0.36:0.64 of γ, Kydland and Prescott (1982)
depreciation rate	δ	0.0177 per quarter	$I/K = 7.1\%$ per year (no growth), the calibration identity of Cooley and Prescott (1995)
Frisch elasticity	$1/\nu$	1	top of the microeconometric range, below the values RBC models use, King and Rebelo (1999)
steady-state hours	N_{ss}	1/3	standard time-use value, Cooley and Prescott (1995). Sets the scale χ
TFP persistence	ρ_z	0.95	Cooley and Prescott (1995), Kydland and Prescott (1982)
wage persistence	ϕ_w	0.833	base-wage duration of six quarters, Grigsby et al. (2021). Appendix B

NOTES: Quarterly values. The hours level is a choice of units, since replacing N by cN and χ by $\chi c^{-(1+\nu)}$ leaves the GHH period utility and every reported moment unchanged, so $N_{ss} = 1/3$ sets χ and restricts nothing. The wage persistence is set from external evidence on wage durations rather than from a steady-state moment, so it is listed apart. Its conversion from a duration to ϕ_w is in Appendix B. ρ and σ_z are calibrated internally (Table 2).

is reported, and at six quarters gives $\phi_w = 0.833$, a half-life of 0.95 yr, the baseline. Read as a fixed-length staggered contract rather than a constant hazard, the same six quarters map to a shorter real half-life of 0.75 yr, $\phi_w = 0.794$ (Appendix B), so the assigned interval is 0.75–0.95 yr, over which Subsection 6.6 re-solves the model. The stickiness the duration measures is that of the real factor price, the Dunlop–Tarshis acyclicality of the aggregate real wage, and Hall (2005) supplies its adjustment form.

4.2 Internally calibrated parameters

Three parameters are calibrated inside the full nonlinear model to three level or steady-state targets, listed in Table 2. With ϕ_w assigned, only these three are free. (ϕ_r, ρ) are solved jointly on the full model and σ_z is chosen after every other parameter is set.

Table 2: Internally calibrated parameters.

description	parameter	value	moment	data	model
rental persistence	ϕ_r	0.571	binding-constraint ratio, labor- to capital-limited plants, US Census QPC	1.54	1.55
discount rate	ρ	0.0106 per quarter	capital–output ratio K/Y, annual	2.85	2.85
TFP innovation	σ_z	0.00309	std($\log Y$), HP(1600)	0.0163	0.0163

NOTES: The ergodic K/Y and the gross investment share I/Y are one moment, since $I = \delta k$. σ_z is re-fitted last under the Kydland–Prescott convention at every parameter point, so std($\log Y$) is matched by construction and is not a test of the model.

Binding-constraint ratio. The rental persistence ϕ_r matches the binding-constraint ratio. In the US Census Quarterly Survey of Plant Capacity Utilization a plant reports which inputs limit its production, box 54 "insufficient supply of local labor force/skills" and box 56 "equipment limitations." A plant may mark several reasons, the thirteen boxes summing to about 170 percent in 2020Q4, so each share is a marginal probability. The model counterpart is the ergodic fraction of time the representative firm's labor constraint binds, $\Pr(\text{labor binds}) = \Pr(\text{reg}_N) + \Pr(\text{reg}_{KN})$, against $\Pr(\text{capital binds}) = \Pr(\text{reg}_K) + \Pr(\text{reg}_{KN})$, the plant shares read as draws from the stationary distribution, and only the ratio is targeted.

The ratio pins the ratio of the two adjustment speeds rather than either level,[6] the rental adjusting about 3.0 times faster than the wage, a fitted rental half-life of 0.31 yr at the assigned wage half-life of 0.95 yr, and Subsection 6.6 follows that factor across the wage range.[7]

Capital–output ratio and output volatility. The discount rate ρ matches the ergodic capital–output ratio K/Y, equivalently the gross investment share I/Y. The innovation size σ_z matches output volatility std($\log Y$) = 0.0163 (HP(1600)) under the Kydland–Prescott convention, refit whenever another parameter changes, so std($\log Y$) is matched by construction.[8] The frictionless economy of Section 6 is calibrated to the same target with its own innovation size, $\sigma_z = 0.0090$ against 0.0031 with the friction, the friction supplying the dif-

[6]The levels are not comparable. A box marks a constraint that holds the plant below capacity, the model counts every quarter in which a ceiling binds at all, and the shares, 13 and 8 percent of plants, sit well below the model's 0.47 and 0.30.

[7]The slow adjustment is of the rental price the firm faces, not the plant-level capital stock. ϕ_r governs how fast the factor price returns to its competitive value, and is silent about the lumpiness of individual investment episodes.

[8]The small innovation is consistent with measured productivity, because idle capital loads onto a Solow residual computed from the owned stock. Measured as in the data, with shares 0.36 and 0.64, the model's residual has a cyclical standard deviation of 0.0068 against 0.0085 in the data, while true productivity has 0.0040. Goods-market search moves measured productivity the same way, there from a demand shock (Bai et al., 2026).

ference. Within the fitted block ρ moves the level of K/Y and ϕ_r the regime shares, so the two are separately identified.

The three targets are hit, the binding-constraint ratio at 1.55 against 1.54, the annual K/Y at 2.85 against 2.85, and std($\log Y$) at 0.0163 by construction, and the sampling error of the survey ratio is in the target record of Appendix OA.3.2. The ergodic regime shares are $(\Pr(\text{reg}_\emptyset), \Pr(\text{reg}_K), \Pr(\text{reg}_N), \Pr(\text{reg}_{KN})) = (0.481, 0.050, 0.217, 0.252)$, giving Pr(labor binds) = 0.469 and Pr(capital binds) = 0.302, whose ratio is the matched 1.55. Three provisos qualify the binding-constraint target, each given in full in Appendix OA.3. Only std($\log Y$), K/Y, and the binding-constraint ratio are targets, and every other moment in Section 6 is untargeted. Figure 1 maps the four regimes in the price-gap plane with the ergodic distribution the ratio is computed on. The economy visits all four, so rationing is a routine phase of the cycle and not a rarity.

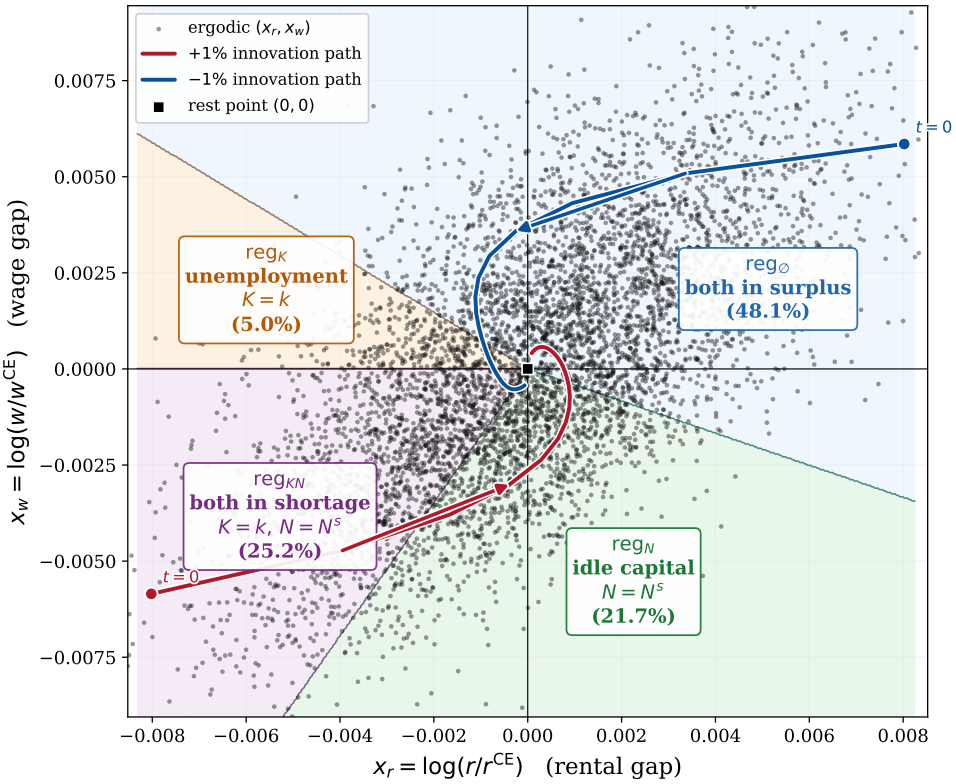


Figure 1: The four KKT regimes in the price-gap plane (x_r, x_w), with $x_p \equiv \log(p/p^{\text{CE}})$, colored by which ceilings bind, the regimes $\text{reg}_\emptyset, \text{reg}_K, \text{reg}_N, \text{reg}_{KN}$ of Subsection 3.2, with the ergodic distribution and the two isolated ± 1 percent impulse paths.

NOTES: Dark points are the realized ergodic distribution of (x_r, x_w), the regime shares are labeled in the plot, and the axes span the 0.5th–99.5th percentiles of the simulated cloud. The red and blue curves are the two isolated impulse transitions from the rest point at the origin, one $+1$ percent and one -1 percent innovation to $\log z$ at $t = 0$ with no further innovations and the household on its equilibrium policy. An adverse innovation opens both gaps upward and puts the economy in the both-in-surplus regime, where idle capital and unemployment coexist, and a favorable one opens both downward and puts it at the both-in-shortage corner. As the faster rental gap closes first, the bust passes through unemployment alone and the boom through idle capital alone on the way back to the origin.

5 The mechanism

This section follows a productivity innovation through the two price gaps on the calibrated model to show why busts run deeper than booms.

5.1 A shock and the two price gaps

Let ε be a one-percent innovation to log productivity that hits the economy at its rest point, the state it settles to in the absence of innovations, both prices at their clearing values and the capital stock at its rest-point level. Under Cobb–Douglas technology the clearing prices answer the innovation exactly log-linearly, the rental by $a_r = 1 + \beta m = 1.4$ percent and the wage by $a_w = \nu m = 0.7$ percent, $m = 1/(1 + \nu - \beta)$ (Appendix OA.1), while the posted prices cover only the fraction $1 - \phi_p$ of that distance, so the gaps open against the shock,

$$x_r = -\phi_r a_r \, \varepsilon = -0.8 \, \varepsilon, \qquad x_w = -\phi_w a_w \, \varepsilon = -0.6 \, \varepsilon. \tag{18}$$

A favorable innovation leaves both posted prices below their clearing values, an adverse one above, and every impact response below is linear in ε, so the size of the innovation sets scale and not shape. The price rows of Figure 2 show the two transitions, the gaps opening on impact and closing at the two prices' speeds.

5.2 Asymmetric factor elasticities

In the frictionless economy the firm is only a production technology and the two sides of each market move together, employment rising by $m = 0.7$ percent. At sticky posted prices the two sides part. The household's stock cannot change within the quarter, and its hours answer the posted wage with elasticity $1/\nu$. The firm's unconstrained demands K^d and N^d of Subsection 3.2 are exact power functions of the two gaps (Appendix OA.1). Per percent of innovation,

$$\frac{\mathrm{d} \log k}{\mathrm{d}\varepsilon} = 0, \qquad \frac{\mathrm{d} \log N^s}{\mathrm{d}\varepsilon} = \frac{(1 - \phi_w) \, a_w}{\nu}, \tag{19}$$

$$\frac{\mathrm{d} \log K^d}{\mathrm{d}\varepsilon} = \frac{(1 - \beta) \, \phi_r a_r + \beta \, \phi_w a_w}{1 - \gamma}, \qquad \frac{\mathrm{d} \log N^d}{\mathrm{d}\varepsilon} = m + \frac{\alpha \, \phi_r a_r + (1 - \alpha) \, \phi_w a_w}{1 - \gamma}. \tag{20}$$

The household answers a one-percent innovation with a fixed stock and 0.12 percent more hours, and the firm answers it with about 7 percent more of each factor, every demand coefficient carrying the reciprocal pure-profit share $1/(1 - \gamma) = 10$. The supply side is the inelastic side.

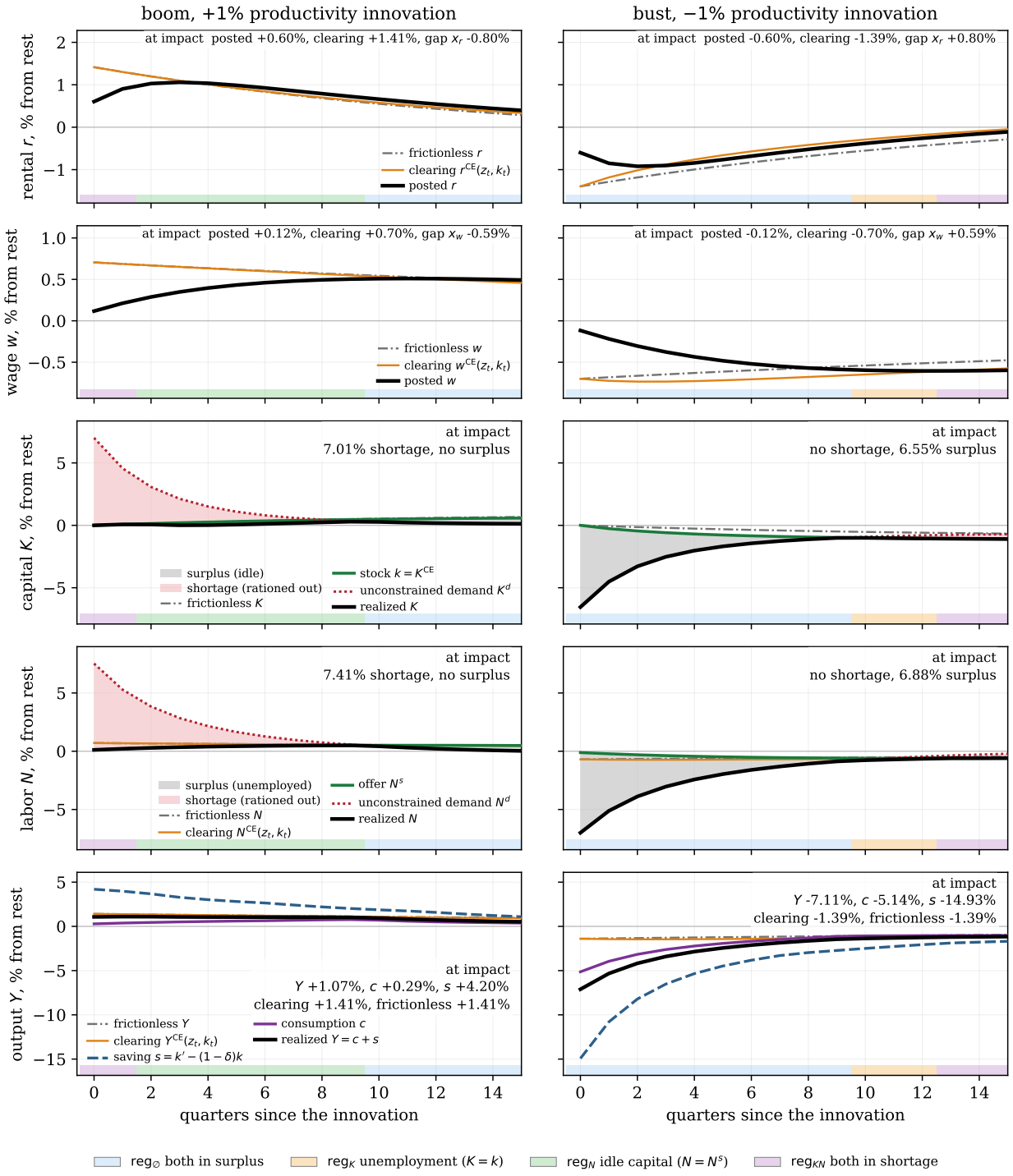


Figure 2: Factor prices, the two rationed factor markets, and output after an isolated one-percent productivity innovation.

NOTES: A ±1 percent innovation to $\log z$ at the rest point, decaying at ρ_z, the household on its stochastic saving policy throughout, with no perfect foresight. Every line is a percent deviation from its own rest point, the frictionless economy on its own transition. In the price rows the distance between the posted price (black) and its clearing value (orange) is the log gap x_r or x_w. In the capital row the clearing value is the stock itself, so the green line carries it, and where labor binds the firm's capital demand is the conditional $K^\dagger$ of Subsection 3.2, below K^d, so realized capital can sit below both the stock and K^d in a boom. The strip marks the regime in the colors of Figure 1.

In the data the firm's side answers the cycle far more than the household's. Measured by the standard deviation of the HP(1600) cyclical component of the quarterly log series, each pair on its common postwar sample, the labor force moves 0.39 as much as employment and 0.05 as much as job vacancies, and the capital stock 0.14 as much as capacity utilization. In the pre-COVID years the three ratios are 0.41, 0.03, and 0.14. The model puts the offer at 0.09 and the stock at 0.13 of the hours and capital the firm employs, the two rows of Table 3 that set the supplied factor against the employed one.

5.3 The short side, and the response by the sign of the shock

In each market the firm employs the smaller of its demand and the ceiling, jointly across the two (Subsection 3.2), so the sign of the innovation decides which side is short and how output answers. After a favorable innovation the demands of (20) run past the supplies of (19), both ceilings bind, and output $zk^{\alpha}(N^{s})^{\beta}$ rises by the innovation plus labor's share of the offer's response. After an adverse innovation both demands fall short, and output $z(K^{d})^{\alpha}(N^{d})^{\beta}$ falls by the innovation plus the two demands' responses weighted by the factor shares. Per percent of innovation,

$$\begin{aligned}
\frac{\mathrm{d}\log Y}{\mathrm{d}\varepsilon}\Big|_{\varepsilon>0} &= \underbrace{1}_{\text{innovation}} + \underbrace{\beta\,\frac{(1-\phi_w)\,a_w}{\nu}}_{\text{household's labor response}} = 1.1, \\
\frac{\mathrm{d}\log Y}{\mathrm{d}\varepsilon}\Big|_{\varepsilon<0} &= \underbrace{1}_{\text{innovation}} + \underbrace{\alpha\,\frac{\mathrm{d}\log K^{d}}{\mathrm{d}\varepsilon} + \beta\,\frac{\mathrm{d}\log N^{d}}{\mathrm{d}\varepsilon}}_{\text{share-weighted firm's inputs' response}} = 7.4,
\end{aligned} \tag{21}$$

against $1+\beta m = 1.4$ for either sign in the frictionless economy. The friction holds the favorable response below the frictionless one and drives the adverse response to 5 times it, so the same innovation moves output 7 times as far when $\varepsilon < 0$. The capital, labor and output rows of Figure 2 show both cases. After the adverse innovation about 7 percent of the stock and of the hours offered are in surplus at once and output falls 7 percent, saving dropping 2 times as far as output and consumption less. After the favorable one nothing is in surplus and output rises little more than the innovation. In the two one-sided regimes one ceiling binds and only the other factor answers the shock. Figure 1 traces the two transitions through the four regimes, and Appendix OA.1 gives each regime's output shortfall in closed form.

Over all quarters output is 2.0 times as volatile in busts as in booms in the model and 1.6 times in the data, so the one-sidedness of the impact responses carries into the size of moves

Table 3: Untargeted business cycle moments in the frictionless economy, the data and the model, and the fraction of the benchmark-to-data gap the friction closes.

moment	frictionless economy	data	model	% of gap closed
Zero or one in the frictionless economy, generated by the friction				
std(u), pp[b]	0	0.95 | 0.79	1.48	**+156**% | **+187**%
corr(u, Y)[b]	0	−0.85 | −0.87	−0.98	**+115**% | **+113**%
$\mathrm{AC}_1(u)$[b]	0	0.75 | 0.90	0.50	**+67**% | **+56**%
skew(u)[b]	0	+2.50 | +0.69	+1.26	**+50**% | **+183**%
cond. sd busts/booms, u[b]	1	2.13 | 1.56	2.34	**+119**% | **+239**%
cond. sd busts/booms, employment N[b]	1	1.62 | 1.15	2.31	**+211**% | **+873**%
cond. sd busts/booms, utilization $\varkappa$[e]	1	1.57 | 1.51	2.33	**+233**% | **+261**%
e.kurt(u)[b]	0	15.88 | 0.36	1.85	**+12**% | **+514**%
std($\log N^s$)/std($\log N$)[b]	1	0.39 | 0.41	0.09	**+149**% | **+154**%
skew($\log Y$)[a]	0	−1.05 | −0.61	−1.10	**+105**% | **+180**%
skew($\Delta \log Y$)[a]	0	−1.04 | −0.22	−0.25	**+24**% | **+114**%
cond. sd busts/booms, Y[a]	1	1.64 | 1.44	2.04	**+163**% | **+236**%
sign-cond. half-life ratio, Y[a]	1	0.66 | 1.50	0.62	**+112**% | **−76**%
e.kurt($\log Y$)[a]	0	3.23 | 0.94	1.51	**+47**% | **+161**%
e.kurt(s)[a]	0	1.34 | 1.20	1.13	**+84**% | **+94**%
corr(LS_t, Y_t), $k = 0$[b]	0	−0.26 | −0.20	−0.44	**+169**% | **+220**%
$\mathrm{AC}_1(\mathrm{LS})$[b]	0	0.60 | 0.62	0.53	**+88**% | **+85**%
std(LS)/std(Y)[b]	0	0.66	0.13	**+20**%
corr(LS_t, Y_{t+k}), data trough k[b]	0	−0.32 | −0.36	−0.22 | −0.13[f]	**+69**% | **+36**%
corr(w, Y)[b]	1	−0.02 | +0.16	0.20	**+78**% | **+95**%
skew($\varkappa$), util.[e]	0	−0.80 | −0.69	−1.30	**+162**% | **+188**%
corr($\varkappa, Y$), util.[e]	0	0.89	0.95	**+107**%
std($\log k$)/std($\log K$), util.[e]	1	0.14 | 0.14	0.13	**+101**% | **+101**%
Present in the frictionless economy, moved toward the data				
std(C)/std(Y)[a]	0.51	0.84 | 0.77	0.70	**+58**% | **+73**%
e.kurt(C)[a]	0.24	12.40 | 0.74	1.61	**+11**% | **+274**%
$\mathrm{AC}_1(I/Y)$[a]	0.71	0.78	0.80	**+129**%
skew(I)[a]	−0.13	−0.69 | −0.63	−1.04	**+163**% | **+182**%
skew(s)[a]	−0.06	−0.51 | −0.48	−0.85	**+176**% | **+188**%
skew(C)[a]	+0.03	−1.57 | +0.13	−1.12	**+72**% | **−1150**%
cond. sd busts/booms, C[a]	0.98	1.29 | 0.88	2.07	**+352**% | **−1090**%
cond. sd busts/booms, I[a]	1.09	1.39 | 1.35	1.87	**+260**% | **+300**%
cond. sd busts/booms, s[a]	1.04	1.25 | 1.22	1.59	**+262**% | **+306**%
e.kurt($\log I$)[a]	0.09	1.46 | 1.22	1.50	**+103**% | **+125**%
No help, or away from the data, each a moment the RBC already fits reasonably				
std(I/Y), cyclical[a]	0.0065	0.0072	0.0062	**−43**%
std(R)/std(Y)[c]	0.12	0.47	0.11	**−3**%
corr(C, Y)[a]	0.99	0.79	1.00	**−5**%
$\mathrm{AC}_1(Y)$[a]	0.71	0.79 | 0.85	0.53	**−225**% | **−129**%
$\mathrm{AC}_1(C)$[c]	0.73	0.65 | 0.84	0.51	**+275**% | **−200**%
cond. sd busts/booms, labor force N^s[b]	1	0.97 | 0.83	0.96	**+133**% | **+24**%
cond. sd busts/booms, capital stock k[e]	1	1.10 | 1.06	1.06	**+60**% | **+100**%

NOTES: [f] The data cells are the correlogram at its trough, $k = +2$ on the full sample and $k = +3$ before 2020, and the two model cells are the model correlogram at those same two lags. The model's own trough is at $k = 0$, where the correlation is −0.44. The last column is $100 \times [\,M_{\text{model}} - M_{\text{RBC}}\,]/[\,M_{\text{data}} - M_{\text{RBC}}\,]$ from the printed cells, 100% the data, 0% the frictionless economy, above 100% an overshoot. Paired cells $a \,|\, b$ are 1947Q1–2026Q1 and 1947–2019. The frictionless column uses its own σ_z, refit to the same output volatility. A frictionless cell within two simulation standard errors of 0 (1 for a ratio) prints 0 (1), and the identities $u \equiv 0$, $\mathrm{LS} \equiv \beta$, $\varkappa \equiv 1$ print 0. Busts are quarters of negative cyclical output and booms the rest. The full-sample kurtosis cells are dominated by 2020, so the 1947–2019 cells are the reliable read. Series, samples, and sources are in Online Appendix Table 7.

inside busts against inside booms, and it sits on the firm's side, employment and utilization 1.6 times as volatile in busts in the data and 2.3 times in the model, while the labor force and the capital stock are symmetric in both, the conditional-volatility rows of Table 3.

5.4 Two speeds

The gaps close at the two prices' own speeds, the rental gap with a half-life of 0.31 yr and the wage gap 0.95 yr, and the two margins recede with them. After an adverse innovation each gap holds back its own factor, so the shortfall rides on both gaps and its rental part, two fifths of it, is gone within a year. After a favorable one the only thing the friction still holds is the offer, which moves with the posted wage, so the cap rides on the wage gap alone and takes three years. A bust therefore fades faster than a boom. The data carry the same sign on 1947Q1–2026Q1, output's sign-conditional half-life ratio, bust to boom, at 0.66 against the model's 0.62, though not on 1947–2019, where it is 1.50. It is the recovery side of the plucking property of Dupraz et al. (2025), a contraction's amplitude forecasting the following expansion's.

5.5 The cost of the friction

The friction's output cost against the frictionless economy has two legs. Over the cycle the static loss at frozen capital, $\log(Y^{\mathrm{CE}}/Y)$, is 1.11 percent. The second leg is the stock. With hours the wealth-free GHH margin, saving is the only intertemporal decision the friction reaches. Only the rented fraction of the stock earns the posted rental, so a unit saved into a bust earns less, and the household holds a mean capital stock 2.14 percent below the frictionless level, its saving rate falling 2.7 times as far after an adverse innovation as it rises after a favorable one. At the ergodic mean the two legs add, with no interaction term, to a 2.09 percent output loss, 0.97 from the stock (Appendix OA.2). In welfare terms the friction costs 1.4 percent of consumption in every state, less than the output loss because the rationed hours are not worked.

Every asymmetry Section 6 sets against the data, in unemployment, in output, in the labor share, and in utilization, comes from this one contrast, a firm that answers a shock with 7 percent of each factor and a household that answers it with almost nothing, and a short side that switches with the sign of the shock.

6 Quantitative scorecard and robustness

The benchmark is the frictionless economy of Subsection 3.5, with its own innovation size refit to the same output volatility. Every moment set against the data is HP(1600) on the logged series, with rates in levels, computed the same way in the model and in the data, and every one is untargeted. The three calibration targets of Section 4 are never tests. Table 3 scores the model on these moments, grouped by what the frictionless economy carries. The per-row record, with series, sample, transform, and identifying object, is Table 7 in Appendix OA.3, and the accuracy of the solution is in Appendix OA.5.3, both in the online appendix. The rows follow the mechanism, unemployment first, then the asymmetric cycle, the labor share, utilization, and the rows the friction does not help. Subsection 6.6 then varies the assigned parameters.

6.1 Involuntary unemployment

Involuntary unemployment is the first match. Cyclical std(u) = 1.48 pp against 0.95 on 1947Q1–2026Q1 and 0.79 on 1947–2019, the two samples used throughout, is exactly zero in the frictionless economy and is generated with no search, matching, or bargaining technology. Its correlation with output is -0.98 against -0.85 and -0.87. The model rate $u = \max(N^s - N, 0)/N^s$ is a within-period rationing rate while the data series is a labor-force rate, so the match is one of order of magnitude, a roughly 1.5× overshoot. Its dynamics match too. The rationing rate is persistent, $\mathrm{AC}_1(u) = 0.50$ against 0.75 and 0.90, and right-skewed, skew(u) = +1.26 against +2.50 and +0.69, rising in sharp jumps and falling slowly. Busts are more volatile than booms, the conditional standard-deviation ratio for unemployment reaching 2.34 against 2.13 and 1.56, each of these zero or one in the frictionless economy.

The search literature reaches unemployment volatility of this scale by making small the fraction of a job's output the market can allocate to job creation, the small fundamental surplus of Ljungqvist and Sargent (2017). My model has no vacancies, so that statistic does not transfer, and it reaches this scale by a different route, on impact the predetermined wage passing none of the productivity innovation into the price, so the short side rations the whole innovation one for one into unemployment (Section 5). Appendix OA.3.5 places this against the fundamental-surplus fraction of Ljungqvist and Sargent (2017).

6.2 Deepness and steepness

A bust-heavy asymmetry runs across every aggregate. Output skewness is -1.10 against -1.05 on 1947Q1–2026Q1 and -0.61 on 1947–2019, the skewness of investment and of saving

are −1.04 and −0.85 against −0.69 and −0.51, that of consumption is −1.12 against −1.57 and +0.13, and the growth-rate skewness is −0.25 against −1.04 and −0.22, each near zero in the frictionless economy. The same rationing thickens both tails, excess kurtosis reaching 1.5 in output, 1.9 in unemployment and 1.1 to 1.5 in investment and saving, near zero in the frictionless economy and beside the 0.9 to 1.2 of 1947–2019, the full-sample figures inflated by the 2020 collapse. Khan and Thomas (2008) find that general equilibrium removes the thick tail that plant-level lumpy investment produces and that annual investment rates show none. The quarterly cyclical series has one, and the friction produces it because the rationing loss is convex in the gaps.

Busts are more volatile than booms throughout, the conditional standard-deviation ratio reaching 2.04 for output against 1.64 and 1.44, one in the frictionless economy, and the investment, saving and consumption rows of Table 3 read the same way against a frictionless economy within a tenth of one. Neftçi (1984) documented unemployment rising in sudden jumps and falling slowly, and Sichel (1993) decomposed the asymmetry into deepness and steepness. Figure 3 plots the ergodic densities of output and unemployment against the data.

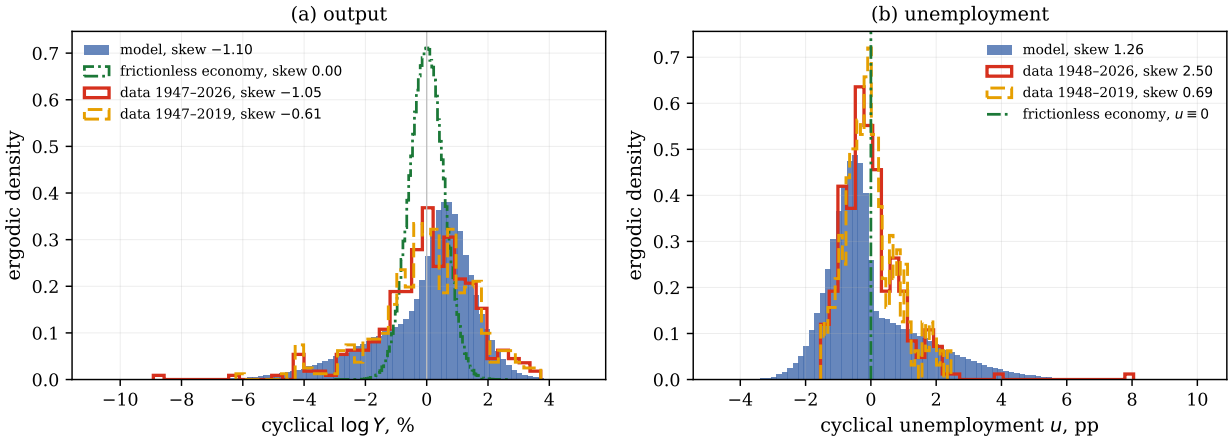


Figure 3: Ergodic densities of cyclical output and unemployment, with the US data density overlaid. NOTES: Model densities from the ergodic simulation, data steps for 1947–2026 and for 1947–2019, each HP(1600)-filtered on its own sample. Panel (a) is cyclical $\log Y$ and panel (b) cyclical involuntary $u = (N^s - N)/N^s$, whose data counterpart is the civilian unemployment rate.

6.3 The labor share

The firm's labor first-order condition $\beta z K^{\alpha} N^{\beta-1} = w$ makes the labor share $\mathrm{LS} \equiv wN/Y = \beta$ exactly whenever labor is interior, so a frictionless economy has $\mathrm{std}(\log \mathrm{LS}) = 0$ and every

movement in the labor share comes from the friction, a property a CES production function would have destroyed.

The model's labor share matches in sign and level, countercyclical at $\mathrm{corr}(\mathrm{LS}, Y) = -0.44$ at $k = 0$ against -0.26 on 1947Q1–2026Q1 and -0.20 on 1947–2019, with a first autocorrelation of 0.53 close to the data 0.60 and 0.62. The share's volatility is small, $\mathrm{std}(\mathrm{LS})/\mathrm{std}(Y) = 0.13$ against 0.66 in the data, because the share moves only when a quantity binds. Panels (b) and (c) of Figure 4 follow it through the two transitions of Figure 2. In the boom, with labor at its ceiling, the share falls on impact by 1.4 times the wage gap, $\log(\mathrm{LS}/\beta) = (1+(1-\beta)/\nu)\, x_w$. In the bust, with both demands interior, the share stays at β until the wage gap changes sign three years on and the ceiling binds. Its timing, however, the model misses. The measured share leads output by two to three quarters and overshoots its own mean about a year after a boom, reaching +0.46 four quarters back (panel (a) of Figure 4). Its output cross-correlation troughs away from zero, -0.32 and -0.36 in the data against -0.22 and -0.13 in the model at $k = +2$ and $k = +3$. Neither a share fixed at β plus a contemporaneous rationing term can produce.

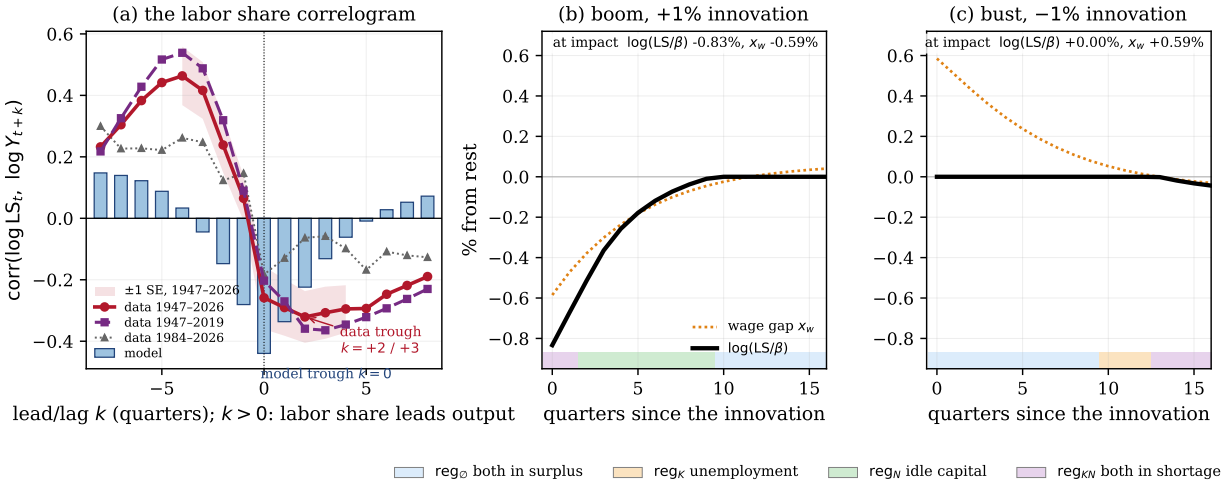


Figure 4: The labor share against US data, its output cross-correlogram in panel (a) and the labor-share and wage gaps on the two transitions of Figure 2 in panels (b) and (c).

NOTES: Panel (a), the cross-correlation $\mathrm{corr}(\log \mathrm{LS}_t, \log Y_{t+k})$, $k > 0$ meaning the labor share leads output. Model bars are the baseline calibration over 300 simulated paths, and data lines are HP(1600) on the logs of wN/Y and Y over 1947–2026, 1947–2019, and 1984–2026, and the shaded band is the full-sample Newey–West ±1 standard error. Panels (b) and (c), $\log(\mathrm{LS}/\beta)$ (black) and the wage gap x_w (orange dotted) on the two transitions of Figure 2, percent from the rest point, with the regime strip in the colors of Figure 1.

6.4 Capacity utilization

The rented fraction cannot exceed one and falls whenever capital is in surplus, so utilization is capped in booms and open in busts. It is procyclical and left-skewed, $\text{corr}(\varkappa, Y) = 0.95$ against 0.89 and $\text{skew}(\varkappa) = -1.30$ against -0.80 on 1948–2024 and -0.69 on 1948–2019, both zero in the frictionless economy. In this model utilization is idle capital under rationing rather than an optimizing choice, so the measured series is a signature check and nothing stronger, and Appendix OA.3.5 separates this idle-capital reading from the optimizing-intensity margin.

6.5 The remaining rows, and the cost in persistence

In 23 rows of Table 3 the frictionless economy is zero or one and the friction generates each with the right sign on 1947Q1–2026Q1, one of them with the opposite sign on 1947–2019. Among them the posted wage's correlation with output, 1 in the frictionless economy, is 0.20 against -0.02 and $+0.16$ in the data, the acyclical real wage of Dunlop and Tarshis. Ten rows that economy already produces move toward the data without being asked, the persistence of the investment rate, the skewness of consumption, investment and saving, their conditional volatility ratios, the relative volatility of consumption, and the kurtosis of consumption and investment. The remaining rows are moments the frictionless economy already fits, the relative volatility of the return, the procyclicality of consumption, the investment-rate volatility, and the symmetry of the labor force and the capital stock over the cycle, on which the friction does not help.

The cost falls on persistence. Output is too weakly autocorrelated, $\text{AC}_1(Y) = 0.53$ in the model against 0.79 on 1947Q1–2026Q1 and 0.85 on 1947–2019 and 0.71 in the frictionless economy, and consumption persistence behaves the same, 0.51 against 0.73 without the friction. The rationing margin re-clears faster than technology decays, so a rationed factor returns toward its clearing quantity within a few quarters and adds a transitory component that lowers the first autocorrelation, while the frictionless economy inherits the quarterly TFP autocorrelation $\rho_z = 0.95$.

6.6 Robustness

No sign in Table 3 changes when the two assigned parameters move across their ranges with the internal parameters recalibrated. Only magnitudes scale. Re-solving the full nonlinear model at both ends and the midpoint of the wage half-life's assigned range 0.75–0.95 yr, with the shock size and the rental speed refit to their targets at each point, moves $\text{std}(u)$

from 1.48 to 1.44 pp, the mean loss from 1.11 to 1.04 percent, and skew(u) from 1.26 to 1.31, so even the fastest admissible wage overshoots the data target near 1.0 pp by about half. What the binding-constraint ratio pins is not the wage level but the ratio of the two speeds, the rental adjusting 3.0 times faster at every point of the range and from 3.0 to 2.4 over the wider 0.50–2.00 yr sweep.

The returns to scale set the size of the loss through the amplifier $1/(1-\gamma)$, not the shape of the cycle. Raising γ from 0.90 to 0.95 lifts std(u) only from 1.48 to 1.54 pp and the mean loss from 1.11 to 1.25 percent, the shock size being refit at each point to hold $\text{std}(\log Y) = 0.0163$, from $\sigma_z = 0.0031$ to 0.0016, which absorbs most of the amplification. Over $\gamma \in [0.85, 0.95]$ the static misallocation leg rises from 0.99 to 1.25 percent, a factor of 1.26 against the 3.0 by which $1/(1-\gamma)$ itself rises, and the total output loss compared to a frictionless economy from 1.77 to 2.49 percent.

7 Conclusion

Two real factor prices that adjust toward their market-clearing values at finite, data-disciplined speeds, and nothing else, hold an otherwise standard real business cycle economy off its markets, so the short side of each rations. Involuntary unemployment follows with no search technology, no matching function, and no nominal rigidity, and a single plant-level moment on the mix of constrained plants pins the one speed the data can identify. It also delivers a countercyclical labor share, a bust-heavy cycle from symmetric shocks, and procyclical, left-skewed utilization, closing most of the frictionless economy's gap on moments it was never asked to fit.

A small stickiness in a relative price, turned by a thin pure-profit share into a quantity response an order of magnitude larger, delivers patterns usually assigned to nominal frictions or to labor-market institutions from mispriced factors alone.

The model has its limits. Unemployment is a stock of idle hours, not a flow, so there is no job-finding rate and no spell duration. The friction costs persistence, output's first autocorrelation falling to 0.53 against 0.71 in the frictionless economy and 0.79 in the data, because a rationed factor returns toward its clearing quantity within a few quarters and adds a transitory component that technology's slow decay does not have, and it leaves the labor share a fifth as volatile as in the data.

The device is portable, since any real price can carry a finite market-clearing speed. Three extensions follow from the model as it stands. A speed that differs between rising and falling prices would give a real, price-based counterpart of the downward wage rigidity the nominal literature imposes directly. Job finding and separation would turn the stock of idle

hours into a flow with a spell length. A nominal goods price of the Calvo or Rotemberg form could sit above the real factor prices and join the friction to a New Keynesian layer.

Appendix A. The price rule, its readings and its alternatives

This appendix derives the rule (17) (Appendix A.1), gives it an optimizing reading (Appendix A.2) and compares its measure of excess demand with two alternatives (Appendix A.3). Throughout, $p_{j,t}^{\mathrm{CE}} \equiv p_j^{\mathrm{CE}}(z_t, k_t)$ for $j \in \{r, w\}$ and $\lambda_j \equiv -\log \phi_j$.

A.1 Derivation of the rule

In logs the unconstrained demands (12) and the labor offer are linear in the log prices, and they clear at $(r^{\mathrm{CE}}, w^{\mathrm{CE}})$ by (13), so the log excess demands at any pair of prices are exactly linear in the gaps,

$$\begin{pmatrix} \log K^d - \log k \\ \log N^d - \log N^s \end{pmatrix} = -\mathcal{J} \begin{pmatrix} x_r \\ x_w \end{pmatrix}, \qquad \mathcal{J} = \frac{1}{1-\gamma} \begin{pmatrix} 1-\beta & \beta \\ \alpha & 1-\alpha+(1-\gamma)/\nu \end{pmatrix}.$$

The wage $\check{w}$ sets the second row to zero, $\log \check{w} - \log w^{\mathrm{CE}} = -(\mathcal{J}_{21}/\mathcal{J}_{22})\, x_r$, and the first row becomes $\mathcal{E}_r = -(\det \mathcal{J}/\mathcal{J}_{22})\, x_r$. The rental $\check{r}$ sets the first row to zero, and the second becomes $\mathcal{E}_w = -(\det \mathcal{J}/\mathcal{J}_{11})\, x_w$. With $\det \mathcal{J} = [1 + (1-\beta)/\nu]/(1-\gamma)$ these are (16), exact for a gap of any size.

Within the quarter (z_t, k_t) is fixed and hours carry no wealth effect, so $p_{j,t}^{\mathrm{CE}}$ is constant, and substituting (16) into (15) gives

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\big(\log p_j(\tau) - \log p_{j,t}^{\mathrm{CE}}\big) = -\omega_j \psi_j \big(\log p_j(\tau) - \log p_{j,t}^{\mathrm{CE}}\big),$$

whose solution at $\tau = 1$ is the rule (17) with $\phi_j = e^{-\omega_j \psi_j}$. The imbalance $\mathcal{E}_j = -\psi_j x_j$ closes at the same speed $\lambda_j = \omega_j \psi_j$. A discrete version, in which each quarter removes the share $1 - \phi_j$ of the imbalance at last quarter's price, $\mathcal{E}_j(\log p_{j,t}) = \phi_j\, \mathcal{E}_j(\log p_{j,t-1}; z_t, k_t)$, gives the same rule.

A.2 The price poster, an optimizing reading

The rule (17) also results from an optimizing intermediary. A short-lived poster on the supply of hours or of capital services inherits last period's post $\log p_{j,t-1}$, observes the clearing target $\log p_{j,t}^{\mathrm{CE}}$, and weighs posting away from it, with curvature $\theta_j > 0$, against changing its post,

with curvature $\zeta_j > 0$, a menu or renegotiation cost,

$$\max_{\log p_{j,t}} \; -\tfrac{\theta_j}{2}\big(\log p_{j,t} - \log p_{j,t}^{\mathrm{CE}}\big)^2 - \tfrac{\zeta_j}{2}\big(\log p_{j,t} - \log p_{j,t-1}\big)^2, \qquad j \in \{r, w\}. \tag{22}$$

The next post is another poster's problem, so only this period's two costs count, and the first-order condition is (17) with $\phi_j = \zeta_j/(\theta_j + \zeta_j)$, the change cost's share of the total curvature. The curvature $\theta_j = -S_j''(\log p_j^{\mathrm{CE}}) > 0$ is that of the transaction surplus $S_j(\log p_j)$ of a supplier of one of many imperfectly substitutable varieties, its revenue net of the disutility of supply, which peaks at the clearing price, and ζ_j is a repricing cost in the sense of Rotemberg (1982). The step from the poster's choice to the rule is exact for any clearing-price process, and the quadratic loss is exact for a quadratic surplus. The adjustment (15) has the form of the myopic limit of Werning and Lorenzoni (2026), in which price setters under a Calvo friction move each price with the current excess demand in its own market, measured there as effective demand, the concept behind the second measure below.

A.3 Two other measures of excess demand

Taking each imbalance after the other market has cleared gives each price an adjustment of its own. Two other measures are natural, each used in (15) in place of (14), with the rest of the model unchanged and the three targets of Table 2 recalibrated. The first takes both markets at the posted prices,

$$\mathcal{E}_r^{\mathrm{P}} \equiv \log K^d(z, r, w) - \log k, \qquad \mathcal{E}_w^{\mathrm{P}} \equiv \log N^d(z, r, w) - \log N^s(w), \tag{23}$$

the rows of $-\mathcal{J}x$ in Appendix A.1, so with $\Omega \equiv \mathrm{diag}(\omega_r, \omega_w)$ a quarter maps the gaps it opens with, $x_t^0 \equiv \log p_{t-1} - \log p_t^{\mathrm{CE}}$, into

$$x_t = e^{-\Omega\mathcal{J}}\, x_t^0. \tag{24}$$

The off-diagonal terms of $\mathcal{J}$ are as large as the diagonal ones, so a wage above clearing pulls the rental below its clearing value.

The second takes the firm's effective demand, its demand in one market given its con-

straint in the other (Benassy, 1975),

$$\mathcal{E}_r^{\mathrm{E}} \equiv \begin{cases} \log K^d(z,r,w) - \log k & \text{if } N^d(z,r,w) \le N^s(w), \\ \log K^{\dagger}\big(z,r,N^s(w)\big) - \log k & \text{otherwise,} \end{cases}$$
$$\mathcal{E}_w^{\mathrm{E}} \equiv \begin{cases} \log N^d(z,r,w) - \log N^s(w) & \text{if } K^d(z,r,w) \le k, \\ \log N^{\dagger}(z,k,w) - \log N^s(w) & \text{otherwise,} \end{cases} \tag{25}$$

with $N^{\dagger}(z,k,w) \equiv (\beta z k^{\alpha}/w)^{1/(1-\beta)}$ the labor demand with capital fixed at k. The measure is continuous, piecewise linear in the gaps and, in a market in surplus, equal to the log of the share of the factor used, and (15) is integrated numerically. In both versions ω_w keeps the wage's response to its own gap at the assigned persistence, $\partial x_{w,t}/\partial x_{w,t}^0\big|_{x_{r,t}^0=0} = \phi_w = 0.833$, averaged over the sign of the gap under effective demand.

At the posted prices the model reaches the three targets with $\omega_r = 0.185$, $\sigma_z = 0.0053$ and $\rho = 0.0104$. The two gaps move in opposite directions, with a correlation of -0.75 against $+0.65$ in the baseline, and the friction amplifies less, so the fit needs a productivity shock 70 percent larger. In Table 4 unemployment volatility and the bust-boom asymmetry of output move toward the data, while the real wage turns procyclical and unemployment and the labor share lose persistence.

Table 4: The baseline and two other measures of excess demand, each calibrated to the targets of Table 2.

moment	data	baseline	at posted prices	effective demand
binding-constraint ratio (target)	1.54	1.55	1.54	0.93
std(u), pp	0.95	1.48	1.32	1.33
corr(u, Y)	−0.85	−0.98	−0.96	−0.94
AC$_1$(u)	0.75	0.50	0.32	0.42
corr(LS$_t, Y_t$)	−0.26	−0.44	−0.47	−0.52
AC$_1$(LS)	0.60	0.53	0.31	0.63
std(LS)/std(Y)	0.66	0.13	0.12	0.22
corr(w, Y)	−0.02	0.20	0.57	0.42
corr($\varkappa, Y$), util.	0.89	0.95	0.67	0.92
skew(log Y)	−1.05	−1.10	−0.97	−1.55
skew($\Delta \log Y$)	−1.04	−0.25	−0.48	−0.44
cond. sd busts/booms, Y	1.64	2.04	1.85	2.43
std(C)/std(Y)	0.84	0.70	0.67	0.64
AC$_1$(Y)	0.79	0.53	0.41	0.52
mean loss $1 - Y/Y^{\mathrm{CE}}$, %		1.11	1.12	0.83

NOTES: Full-sample data as in Table 3. Effective demand at $\omega_r = 0.066$, its value closest to the binding-constraint target, where std(log Y) = 0.0165 and $K/Y = 2.82$.

Under effective demand the binding-constraint ratio stays between 0.84 and 0.93 for ω_r from 0.066 to 0.66, short of the target of 1.54. A market in shortage answers through the

far less elastic constrained demand, so shortages close slowly and both factors are short in more than half of all quarters.

Neither measure built on unconstrained demands tracks the state of a market better than the other. In the baseline economy each signals a capital shortage in one quarter in five in which capital is partly idle, since a firm short of hours cannot use the capital its unconstrained demand calls for. Effective demand avoids this but misses the target. Across the thirteen untargeted rows of Table 4, the baseline column is closest to the data in six rows, the effective-demand column in four and the posted-prices column in three.

Appendix B. From a measured wage duration to the persistence ϕ_w

The wage persistence ϕ_w is assigned from micro evidence, not fitted to a moment (Section 4). The assignment converts a measured wage-spell duration into the persistence of the posted-price rule (17) of Subsection 3.3. The conversion is not assumption-free. It maps the same measured six-quarter duration into a persistence under a stated model of reset timing, and two standard readings of that timing give the two ends of the interval the paper carries. The rule's flexibility ω_w (Subsection 3.3) and the reset hazard h_w used here are distinct parameters.

Take first a constant per-quarter reset hazard h_w, the Calvo reading. It makes the spell between resets geometric, with mean duration the reciprocal of the hazard,

$$\Pr(\text{spell} = j) = h_w(1-h_w)^{j-1}, \qquad \mathbb{E}[j] = \sum_{j\geq 1} j\, h_w(1-h_w)^{j-1} = \frac{1}{h_w}, \qquad j = 1, 2, \ldots$$

The measured base-wage duration of six quarters (Section 4) fixes the hazard, $\mathbb{E}[j] = 6 \Rightarrow h_w = 1/6$. Each quarter a fraction h_w of wages resets to the target (log gap zero) and a fraction $1 - h_w$ carries the previous quarter's gap forward, so the cross-sectional mean of the wage log gap x_w obeys a first-order recursion whose coefficient is the persistence,

$$\mathbb{E}[x_{w,t}] = (1-h_w)\,\mathbb{E}[x_{w,t-1}] \quad \Longrightarrow \quad \phi_w = 1 - h_w = \tfrac{5}{6} = 0.833. \tag{26}$$

The rule is parameterized by the per-quarter speed $-\ln \phi_w$, and a six-quarter geometric spell implies a half-life of the mean gap of $\ln(0.5)/\ln(5/6) = 3.80$ quarters, 0.95 yr, so the per-quarter speed is

$$-\ln \phi_w = \frac{\ln 2}{4 \times 0.95} = 0.182, \qquad \phi_w = 0.833, \tag{27}$$

the value at the top of the assigned interval.

The lower end of the interval is the same six-quarter duration read through fixed-length staggered contracts rather than a constant hazard. If every contract lasts exactly $L = 6$ quarters and a fraction $1/L$ of wages resets each quarter, then after a one-time shock the fraction still carrying the old gap declines linearly, so the cross-sectional mean gap is $\mathbb{E}[x_{w,k}] = (1 - k/6)\,\mathbb{E}[x_{w,0}]$, $k = 0, \ldots, 6$, which reaches half its initial value at $k^\star = 3$ quarters, a half-life of 0.75 yr. Mapping that half-life through the model's geometric rule gives the lower endpoint, $-\ln \phi_w = \ln 2/(4 \times 0.75)$, $\phi_w = 0.5^{1/3} = 0.794$.

The same measured duration therefore delivers $\phi_w = 0.794$ at half-life 0.75 yr under the fixed-length reading and $\phi_w = 0.833$ at half-life 0.95 yr under the constant-hazard reading, the two ends of the assigned interval whose robustness the wage-half-life solves of Section 4 report. The persistence one reads off a duration depends on the reset-timing assumption, so the conversion carries that assumption, the one under which the duration is itself reported.

Two considerations bound the assigned point from below, both toward greater flexibility. The micro durations measure nominal rigidity, whereas ϕ_w is the persistence of the real rented-factor price, and a nominal wage held fixed while the price level drifts is more flexible in real terms, so 0.95 yr is a ceiling. The hazard in these data "first increases and then decreases, with a peak at 12 months" (Barattieri et al., 2014), closer to a fixed-length staggered reset than to the constant-hazard reset of Calvo (1983), so a fixed-length reading maps to a shorter real half-life of 0.75 yr. One consideration points the other way. Posted wages for new hires are rigid downward and flexible upward (Hazell and Taska, 2025), so a symmetric ϕ_w understates the rigidity that binds in a bust.

The constant-hazard equivalence in (26) is exact for the mean gap and only for the mean gap. Calvo resetting disperses prices around that mean, the rule (17) does not, and with one posted price per factor the mean-gap equivalence is the only one that has to hold.

Online Appendix

OA.1 Price gaps, regime losses, and the wedge variance

This appendix records the algebra behind the regime losses (33)–(36) and the return-gap variance (37). The demands and the losses are exact given the two price gaps, because technology is Cobb–Douglas and the labor offer is isoelastic. The gap dynamics and the variance are first-order objects, the competitive prices being linearized in productivity at fixed k. The reported loss and volatility numbers come from the full nonlinear model (Section 4 and Section 5).

OA.1.1 The competitive benchmark in closed form

The competitive system (13) of Subsection 3.3 solves analytically. Equating the two labor conditions gives N^{CE}, and w^{CE} and r^{CE} follow,

$$\begin{aligned} N^{\mathrm{CE}}(z,k) &= \Big(\frac{\beta z k^{\alpha}}{\chi}\Big)^{m}, \quad m \equiv \frac{1}{1+\nu-\beta}, \\ w^{\mathrm{CE}}(z,k) &= \chi\,(N^{\mathrm{CE}}(z,k))^{\nu}, \qquad r^{\mathrm{CE}}(z,k) = \alpha z k^{\alpha-1}(N^{\mathrm{CE}}(z,k))^{\beta}. \end{aligned} \tag{28}$$

The benchmark is analytic at every state, with output $Y^{\mathrm{CE}}(z,k) = z\,k^{\alpha}(N^{\mathrm{CE}}(z,k))^{\beta}$. At the frictionless point ($r = r^{\mathrm{CE}}(z,k)$, $w = w^{\mathrm{CE}}(z,k)$) the firm's unconstrained demand equals both ceilings exactly, so it produces at (k, N^{CE}) with no slack on either margin. Taking logs of the closed form (28) gives the cyclical loadings of the benchmark on the shock,

$$\log N^{\mathrm{CE}} = m\big(\log\beta + \log z + \alpha\log k - \log\chi\big), \qquad a_w \equiv \frac{\mathrm{d}\log w^{\mathrm{CE}}}{\mathrm{d}\log z} = \nu m, \quad a_r \equiv \frac{\mathrm{d}\log r^{\mathrm{CE}}}{\mathrm{d}\log z} = 1+\beta m. \tag{29}$$

These loadings satisfy two identities,

$$a_r - a_w = 1 + (\beta-\nu)m = m = \frac{1}{1+\nu-\beta} > 0, \qquad a_r = \Big(1+\tfrac{1}{\nu}\Big)a_w. \tag{30}$$

The rental is strictly more procyclical than the wage at the efficient point, since $a_r - a_w = m > 0$.

OA.1.2 The unconstrained interior demands

Let $x_p \equiv \log(p/p^{\mathrm{CE}})$ be the price gaps and $n \equiv \log(N/N^{\mathrm{CE}})$, with $\log\varkappa$ the log of the rented fraction $\varkappa \equiv K/k$. In logs the two interior first-order conditions of Appendix OA.4, less their competitive counterparts at the same k, read exactly

$$(\alpha-1)\,\log\varkappa + \beta\,n = x_r, \qquad \alpha\,\log\varkappa + (\beta-1)\,n = x_w,$$

a linear system with determinant $(\alpha-1)(\beta-1) - \alpha\beta = 1-\gamma > 0$. Inverting,

$$\log\varkappa^{d} = \frac{(\beta-1)\,x_r - \beta\,x_w}{1-\gamma}, \qquad n^{d} = \frac{-\alpha\,x_r + (\alpha-1)\,x_w}{1-\gamma},$$

the log-deviations of the unconstrained demands that the KKT conditions test the ceilings against.

OA.1.3 The regime partition and the conditional demand

The two price gaps alone fix the regime, and the four partition the plane as

$$\begin{aligned} \text{reg}_\emptyset &: \; x_r > -\tfrac{\beta}{1-\beta}\, x_w \text{ and } \alpha x_r > -\tfrac{\Theta}{\nu}\, x_w, & \qquad \text{reg}_K &: \; x_w > 0 \text{ and } x_r \le -\tfrac{\beta}{1-\beta}\, x_w, \\ \text{reg}_N &: \; x_r > \tfrac{\beta}{\nu}\, x_w \text{ and } \alpha x_r \le -\tfrac{\Theta}{\nu}\, x_w, & \qquad \text{reg}_{KN} &: \; x_r \le \tfrac{\beta}{\nu}\, x_w \text{ and } x_w \le 0, \end{aligned} \tag{31}$$

with $\Theta \equiv \nu(1-\alpha)+(1-\beta)(1-\alpha)-\alpha\beta = 0.776$ at the baseline calibration, the four boundaries straight lines through the origin, so the regimes are cones meeting at the competitive point. When labor binds the firm's capital demand is the conditional demand $K^\dagger(z, r, N^s)$ of (11),

$$K = \min\{K^\dagger(z, r, N^s),\, k\} \quad \text{whenever labor binds,} \tag{32}$$

which lies below the unconstrained K^d because capital and labor are complements and the firm cannot get the labor to work the extra capital.

OA.1.4 The four regime losses

Each loss is $\mathcal{L} \equiv \log(Y^{\text{CE}}/Y)$ evaluated at the regime's realized (K, N), with the unconstrained demands above testing which ceilings bind. The four are

$$\text{regime reg}_\emptyset \text{ (neither ceiling binds)}: \quad \mathcal{L}_\emptyset = \frac{\alpha\, x_r + \beta\, x_w}{1-\gamma}, \tag{33}$$

$$\text{regime reg}_K \ (K = k, \text{ unemployment}): \quad \mathcal{L}_K = \frac{\beta}{1-\beta}\, x_w, \tag{34}$$

$$\text{regime reg}_N \ (N = N^s, \text{ idle capital}): \quad \mathcal{L}_N = \frac{\alpha\, x_r - (\beta/\nu)\, x_w}{1-\alpha}, \tag{35}$$

$$\text{regime reg}_{KN} \ (K = k,\, N = N^s, \text{ both-bind}): \quad \mathcal{L}_{KN} = -\frac{\beta}{\nu}\, x_w. \tag{36}$$

with the amplifier $1/(1-\gamma)$ on the both-in-surplus regime and the household's own elasticity $1/\nu$ on the both-bind corner, the algebra of Section 5 in the price gaps rather than the innovation. In the *both-in-surplus* regime $\text{reg}_\emptyset$ both ceilings are slack, the firm hires its unconstrained demands, and the output gap at fixed k is $\log Y - \log Y^{\text{CE}} = \alpha \log \varkappa^d + \beta\, n^d$. Substituting the inverted demands and using $\alpha + \beta = \gamma$,

$$\mathcal{L}_\emptyset = -\big(\alpha \log \varkappa^d + \beta\, n^d\big) = \frac{\alpha\, x_r + \beta\, x_w}{1-\gamma},$$

which is (33), the amplification $1/(1-\gamma)$ being the inverse of the determinant $1-\gamma$ of the demand system inverted above.

In the *unemployment* regime reg_K, capital is fixed at $K = k$ and labor sits on its own first-order condition $N = (\beta z k^\alpha / w)^{1/(1-\beta)}$, while the competitive labor solves the same condition at w^{CE}. Since output at fixed k moves only through labor, $\mathcal{L}_K = \beta\,(\log N^{\text{CE}} - \log N)$, and the two first-order conditions differ only in the wage, $\log N^{\text{CE}} - \log N = (\log w - \log w^{\text{CE}})/(1-\beta) = x_w/(1-\beta)$. Hence

$$\mathcal{L}_K = \frac{\beta}{1-\beta}\, x_w,$$

which is (34), and the rental gap is irrelevant once $K = k$ is fixed. The two price-gap conditions of (31) follow the same way. Labor is slack when the labor re-optimized at $K = k$ falls below the offer, $N^\dagger(k) < N^s$, which by $\log N^\dagger(k) - \log N^{\text{CE}} = -x_w/(1-\beta)$ against $\log N^s - \log N^{\text{CE}} = x_w/\nu$ holds exactly when $x_w > 0$. Capital binds when the unconstrained capital demand weakly exceeds the stock, $K^d \geq k$, which by the inverted demands $\log K^d - \log k = [(\beta-1)x_r - \beta x_w]/(1-\gamma)$ is $x_r \leq -\frac{\beta}{1-\beta} x_w$.

In the *idle-capital* regime reg_N, labor is at its ceiling $N = N^s = (w/\chi)^{1/\nu}$ and capital is re-optimized against the rental, $K = (\alpha z (N^s)^\beta / r)^{1/(1-\alpha)}$; substituting this K into $Y = zK^\alpha (N^s)^\beta$, comparing with $Y^{\text{CE}} = zk^\alpha (N^{\text{CE}})^\beta$, and using $\log N^s - \log N^{\text{CE}} = x_w/\nu$ (the GHH ceiling loads on the wage gap with slope $1/\nu$) collects to

$$\mathcal{L}_N = \frac{\alpha\, x_r - (\beta/\nu)\, x_w}{1-\alpha},$$

which is (35). Its two conditions of (31) follow likewise. Capital is slack when the capital re-optimized at $N = N^s$ falls below the stock, $K^\dagger(N^s) < k$, which by $\log K^\dagger(N^s) - \log k = [(\beta/\nu)x_w - x_r]/(1-\alpha)$ holds exactly when $x_r > (\beta/\nu)x_w$, with no α because at $N = N^s$ capital alone adjusts. Labor binds when the labor demanded at that capital weakly exceeds the offer N^s, which carrying the loadings through the labor first-order condition is $\alpha x_r \leq -(\Theta/\nu)x_w$ with $\Theta \equiv \nu(1-\alpha) + (1-\beta)(1-\alpha) - \alpha\beta$.

At the *both-bind* corner reg_{KN} no demand is inverted at all, since both factors sit at their ceilings $K = k$ and $N = N^s$. Output is $Y = zk^\alpha (N^s)^\beta$ against $Y^{\text{CE}} = zk^\alpha (N^{\text{CE}})^\beta$, the common zk^α cancels, and with the GHH offer $\log N^s - \log N^{\text{CE}} = x_w/\nu$ the loss collapses to

$$\mathcal{L}_{KN} = -\frac{\beta}{\nu}\, x_w,$$

which is (36), the rental gap being inert there because capital cannot move off its ceiling.

OA.1.5 The gap dynamics and the return-gap variance

The model is in discrete time, one quarter to the period, so the return-gap variance is read off the model's difference equation. Start from the adjustment law (17), in discrete form,

$$\log p_t = \phi_p \log p_{t-1} + (1-\phi_p) \log p^{\mathrm{CE}}(z_t, k_t), \qquad \phi_p \in (0,1), \quad p \in \{r, w\},$$

in which the posted price partially adjusts to the contemporaneous competitive benchmark. Hold capital fixed and write $\log p^{\mathrm{CE}} = a_p \log z + \text{const}$ (loadings a_r, a_w from (29)). Subtracting $\log p_t^{\mathrm{CE}}$ from both sides and using $x_{p,t} \equiv \log p_t - \log p_t^{\mathrm{CE}}$,

$$x_{p,t} = \phi_p\big(x_{p,t-1} + \log p_{t-1}^{\mathrm{CE}} - \log p_t^{\mathrm{CE}}\big) = \phi_p\, x_{p,t-1} - \phi_p a_p\,(\log z_t - \log z_{t-1}).$$

The forcing is the discrete AR(1) $\log z_t = \rho_z \log z_{t-1} + \sigma_z \varepsilon_t$, $\varepsilon_t \sim N(0,1)$, so $\log z_t - \log z_{t-1} = (\rho_z - 1)\log z_{t-1} + \sigma_z \varepsilon_t$ and the gap obeys

$$x_{p,t} = \phi_p\, x_{p,t-1} - \phi_p a_p\big[(\rho_z - 1)\log z_{t-1} + \sigma_z \varepsilon_t\big].$$

Stacking the state $s_t \equiv (\log z_t,\ x_{p,t})'$ gives the two-state linear Gaussian system

$$\begin{pmatrix} \log z_t \\ x_{p,t} \end{pmatrix} = \underbrace{\begin{pmatrix} \rho_z & 0 \\ -\phi_p a_p(\rho_z - 1) & \phi_p \end{pmatrix}}_{A_p} \begin{pmatrix} \log z_{t-1} \\ x_{p,t-1} \end{pmatrix} + \underbrace{\begin{pmatrix} 1 \\ -\phi_p a_p \end{pmatrix}}_{b_p} \sigma_z \varepsilon_t.$$

The eigenvalues of A_p are ρ_z and ϕ_p, both strictly inside the unit circle for any positive speed, so the system is stationary, and this is what stops sticky prices from destabilizing the model. The stationary covariance $\Sigma_p \equiv \mathbb{E}[s_t s_t']$ is the solution of the discrete Lyapunov equation

$$\Sigma_p = A_p\, \Sigma_p\, A_p' + \sigma_z^2\, b_p b_p', \tag{37}$$

and $\mathrm{Var}(x_p)$ is its lower-right entry, the log-linear closed form for the return-gap dispersion. Solving it at the baseline calibration ($\phi_r = 0.571$, $\phi_w = 0.833$, $\rho_z = 0.95$, $\sigma_z = 0.00309$, $a_r = 1.40$, $a_w = 0.702$) gives

$$\mathrm{std}(x_r) = 0.0029625, \qquad \mathrm{std}(x_w) = 0.0029637,$$

against the full nonlinear model's $\mathrm{std}(x_r) = 0.002925$ and $\mathrm{std}(x_w) = 0.003269$. The discrete log-linear formula therefore overstates the nonlinear return-gap dispersion by about 1.3%, and the discrete form is the reported object.

The return gap that drives the intertemporal channel depends on the rental speed alone among the two adjustment speeds.

Equal rental and wage speeds do not remove the static loss. The two competitive prices load unequally on productivity, so the posted factor-price ratio still departs from the marginal rate of technical substitution, and a relative-price wedge worth about 0.12 to 0.20 percent of output survives at any common finite speed.

The ergodic densities of the two price gaps are in Figure 5.

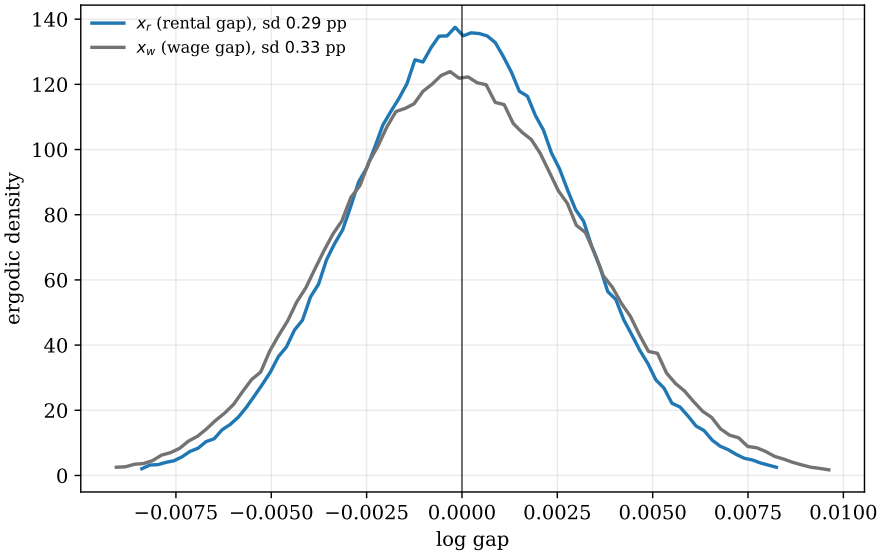


Figure 5: Ergodic densities of the two price gaps x_r, x_w at the baseline calibration.

NOTES: The baseline simulation of the full nonlinear model, $300 \times 1{,}000$ ergodic quarters. Each density is a histogram over the 0.2nd to 99.8th percentiles of its series. The log-linear closed form for the return-gap dispersion $\text{std}(x_r)$ is the discrete Lyapunov equation (37) in Appendix OA.1.

OA.2 The intertemporal distortion

The household saves against the return an owned unit of capital earns, and that return departs from its frictionless level for two reasons. The posted rental is stuck away from competitive, so even a fully rented unit is mispriced, and only the rented fraction earns the rental, so an idle unit earns nothing yet still depreciates at δ. The first is a sticky-rental distortion, the second an idle-capital distortion, the capital-side reading of the same rental gap and idle fraction that drove the within-period distortion. The two shape the level of the capital stock and the asymmetry of saving, taken in turn.

OA.2.1 The level of capital

The household's gross realized return on an owned unit of capital enters its Euler equation (6) as

$$R'(z', k', x'_r, x'_w) \;=\; 1 - \delta + r'(z', k', x'_r)\, \varkappa'(z', k', x'_r, x'_w), \qquad \varkappa' = \frac{K'}{k'} \in (0, 1], \tag{38}$$

in which only the rented fraction $\varkappa'$ earns the sticky rental r' and the idle remainder $1 - \varkappa'$ still depreciates at δ. A prime denotes a next-period value, suppressed in the text hereafter. Add and subtract the competitive rental on the rented fraction to split the gap between this return and its frictionless value $R'^{\mathrm{CE}} = 1 - \delta + r^{\mathrm{CE}}$ exactly,

$$R' - R'^{\mathrm{CE}} = \underbrace{\varkappa' \left(r' - r^{\mathrm{CE}}\right)}_{\text{sticky-rental}} + \underbrace{r^{\mathrm{CE}} \left(\varkappa' - 1\right)}_{\text{idle-capital}}. \tag{39}$$

Which part moves the mean capital stock is settled by the Euler equation averaged over the ergodic distribution. With $\tilde{c}'$ the GHH composite $c - \chi(N)^{1+\nu}/(1+\nu)$ of Subsection 3.1 and ρ the rate of time preference, the expectation of (6) and the law of iterated expectations give the exact restriction

$$1 + \rho = \mathbb{E}[R'] + \frac{\mathrm{Cov}(R', 1/\tilde{c}')}{\mathbb{E}[1/\tilde{c}']} \equiv \mathbb{E}[R'] + \Pi, \tag{40}$$

in which the precautionary premium Π is negative when the realized return is low in the states where marginal utility is high. The frictionless economy obeys the same restriction with $\varkappa' \equiv 1$, its own competitive rental and premium Π^{CE}. Define the mean rental-income gap between the two economies, per owned unit,

$$S \;\equiv\; \mathbb{E}\big[r^{\mathrm{CE}}(z', k'^{\mathrm{CE}})\big] - \mathbb{E}\big[r'(z', k', x'_r)\, \varkappa'(z', k', x'_r, x'_w)\big]. \tag{41}$$

Subtracting the frictional restriction from the frictionless one and canceling the common $1 - \delta$ gives $S = \Delta\Pi$, with $\Delta\Pi \equiv \Pi - \Pi^{\mathrm{CE}}$,

$$S = \Delta\Pi, \qquad \Delta\Pi \equiv \Pi - \Pi^{\mathrm{CE}}. \tag{42}$$

Equation (42) is an identity, and it is not the statement that precaution explains the gap. Each economy's household saves until the mean return an owned unit earns equals $1 + \rho$ less its own small precautionary premium, so the mean return per owned unit is the same in the two economies. The friction lowers what a unit earns, since idle capital earns nothing and

the sticky rental misprices the rest, so for the mean return to be equal the competitive rental must be higher in the frictional economy, which happens only if its capital stock is smaller.

The gap S telescopes into three differences of ergodic means, $S = A + M + U$. The idle-capital term $U \equiv \mathbb{E}[r'(1-\varkappa')]$ is the rental forgone on capital not rented, the rental-mispricing term $M \equiv \mathbb{E}[r^{\mathrm{CE}}(z', k')] - \mathbb{E}[r']$ is the return lost when the posted rental sits away from competitive at the frictional stock, and the schedule term $A \equiv \mathbb{E}[r^{\mathrm{CE}}(z', k'^{\mathrm{CE}})] - \mathbb{E}[r^{\mathrm{CE}}(z', k')]$ is the rise in the competitive rental produced by a lower stock. The schedule term moves with mean log capital through the elasticity $\eta \equiv -\,\mathrm{d}\log r^{\mathrm{CE}}/\mathrm{d}\log k = 1 - \alpha a_r = 0.545$, up to a curvature remainder $J_A \equiv A - r^{\mathrm{CE}}\eta\,\Delta\log\bar{k}$ worth $+0.02\%$ of mean capital, second-order. Putting $S = \Delta\Pi$ into $S = A + M + U$ and solving for $\Delta\log\bar{k} \equiv \mathbb{E}[\log k] - \mathbb{E}[\log k^{\mathrm{CE}}]$,

$$\Delta\log\bar{k} \;=\; \frac{\Delta\Pi - U - M - J_A}{r^{\mathrm{CE}}\,\eta}. \tag{43}$$

Every term reads off the solved ergodic distribution, with σ_z held fixed across the two economies. Table 5 reports the terms at the baseline calibration.

Table 5: The intertemporal channel, the fall in mean capital by term of Equation (43).

term	definition	$\Delta\log\bar{k}$
idle capital U	$\mathbb{E}[r'(1-\varkappa')]$	-2.15%
rental mispricing M	$\mathbb{E}[r^{\mathrm{CE}}(z', k')] - \mathbb{E}[r']$	-0.00%
excess precautionary premium $\Delta\Pi$	$\Pi - \Pi^{\mathrm{CE}}$	-0.00%
schedule curvature J_A	$A - r^{\mathrm{CE}}\eta\,\Delta\log\bar{k}$	$+0.02\%$
total	Equation (43)	-2.14%

NOTES: Each term is a difference of ergodic means of the full nonlinear model at the baseline calibration, 400,000 ergodic states, with σ_z fixed at its baseline value in both economies. The common denominator is $r^{\mathrm{CE}}\eta = (\rho+\delta)(1-\alpha a_r)$, with $\eta = 0.545$. The curvature J_A collects the convexity of diminishing returns, the two economies' different dispersion of capital and the anchoring of the linear term at $\rho+\delta$, together $+0.02\%$ of mean capital.

One term carries the channel. The idle-capital term U is worth 2.15% of mean capital, the mispricing M nets to nothing, since the rental gap has a zero log-mean under the rule (17), and $\Delta\Pi$ and J_A are second-order. Mean capital therefore sits 2.13% below its frictionless level, all of it idle depreciation, not precaution against a more volatile return. A rented-fraction-naive economy that prices every owned unit as though it earned the rental carries no idle-capital term, and its mean capital sits 2.46% above the true economy's, so the whole fall is the price of capital owned but not rented. Both U and the fall rise with the rental persistence ϕ_r, while the wage speed does not enter the Euler equation.

OA.2.2 The asymmetry of saving over the cycle

The two parts of the return wedge carry opposite cyclical signatures. The sticky-rental part is countercyclical, the posted rental lagging the clearing rental, while the idle-capital part lowers the return only when capital is idle, after an adverse innovation. On the two transitions of Figure 2 the saving rate rises 0.6 percentage points after the favorable innovation and falls 1.7 after the adverse one, 2.7 times as far, against a symmetric 0.6 either way in the frictionless economy. The wedge behind the fall is the idle-capital part, 18 basis points per quarter on impact against a sticky-rental part of 2 with the opposite sign. Figure 6 shows consumption falling 5 percent after the adverse innovation against 0.7 without the friction and the hours offer moving a sixth as much as frictionless hours either way. Figure 7 shows the saving rate and the two parts of the wedge.

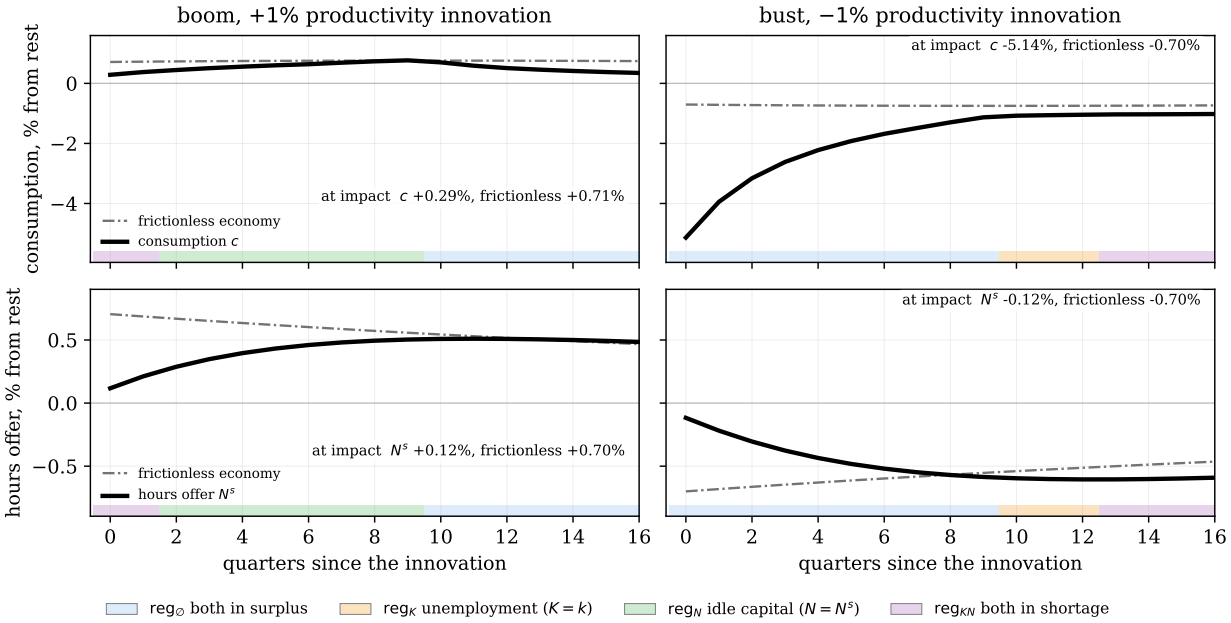


Figure 6: Consumption and the hours offer after an isolated one-percent productivity innovation.
NOTES: The two transitions of Figure 2, every line a percent deviation from its own economy's rest point, the friction economy in black and the frictionless economy on its own transition in gray. The strip marks the regime in the colors of Figure 1.

The same asymmetry appears in the data's cyclical moments (Table 3).

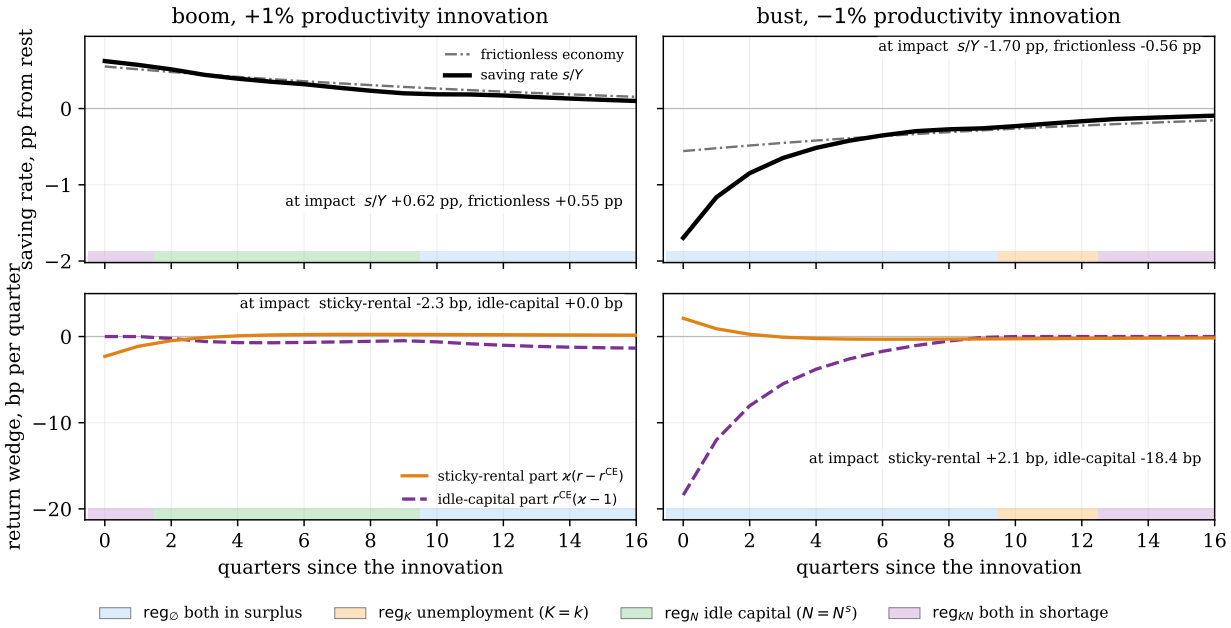


Figure 7: The saving rate and the two parts of the household's return wedge after an isolated one-percent productivity innovation.

NOTES: The two transitions of Figure 2. Top row, the saving rate s/Y in percentage points from each economy's rest point, the friction economy in black and the frictionless economy in gray. Bottom row, the wedge between the rental return the household earns, $\varkappa r$, and the clearing rental, split into the sticky-rental part $\varkappa(r - r^{\mathrm{CE}})$ and the idle-capital part $r^{\mathrm{CE}}(\varkappa - 1)$, in basis points per quarter. The strip marks the regime in the colors of Figure 1.

OA.2.3 The two legs of the cost

Both legs of the friction, the within-period misallocation and the across-period fall in capital, add at the ergodic mean with no interaction term. Sizing them side by side gives the following.

At the baseline calibration the full nonlinear model gives a mean static loss of $\bar{\mathcal{L}} = 1.13$ percent in the log units of Equation (44) and 1.11 percent in the level convention $1 - Y/Y^{\mathrm{CE}}$.

At the ergodic mean the two margins add exactly, with no interaction term. Compare the frictional economy at its own ergodic distribution with the frictionless economy at its own, along a shared productivity path so that the z-contribution cancels. Write $k^* = \mathbb{E}_{\mathrm{fric}}[k]$ and $k^{\mathrm{CE}} = \mathbb{E}_{\mathrm{frictionless}}[k]$, and add and subtract the frictionless output evaluated at the frictional

capital, a telescoping identity

$$\begin{aligned}\Delta Y &\equiv \underbrace{\mathbb{E}\big[\log Y^{\mathrm{CE}}(z,k^{\mathrm{CE}})\big]}_{\text{frictionless}} - \underbrace{\mathbb{E}\big[\log Y(z,k^{*},x_r,x_w)\big]}_{\text{frictional}} \\ &= \underbrace{\Big\{\mathbb{E}[\log Y^{\mathrm{CE}}(z,k^{\mathrm{CE}})] - \mathbb{E}[\log Y^{\mathrm{CE}}(z,k^{*})]\Big\}}_{\Delta_{\mathrm{INT}}=\text{intertemporal}} + \underbrace{\Big\{\mathbb{E}[\log Y^{\mathrm{CE}}(z,k^{*})] - \mathbb{E}[\log Y(z,k^{*},x_r,x_w)]\Big\}}_{\Delta_{\mathrm{STAT}}=\text{static}}.\end{aligned} \tag{44}$$

The static leg is the misallocation loss with capital held at the frictional mean, $\Delta_{\mathrm{STAT}} = \bar{\mathcal{L}}$, and the intertemporal leg is the pure capital-stock move fed through the frictionless technology,

$$\Delta_{\mathrm{INT}} \;\approx\; \alpha\, a_r \left(\mathbb{E}[\log k^{\mathrm{CE}}] - \mathbb{E}[\log k^{*}]\right), \qquad \alpha\, a_r = \alpha(1+\beta m) = 0.455, \tag{45}$$

using the frictionless output elasticity $\mathrm{d}\log Y^{\mathrm{CE}}/\mathrm{d}\log k = \alpha\, a_r$ from the loadings (30). The ordering term that this split could leave behind vanishes identically under Cobb–Douglas, because the misallocation loss $\log(Y^{\mathrm{CE}}/Y)$ is scale-free in the capital level.[9]

At the baseline calibration,[10] the two legs and their sum are

$$\underbrace{\Delta_{\mathrm{STAT}} = 1.13\%}_{\text{static, } k \text{ frozen}} + \underbrace{\Delta_{\mathrm{INT}} = 0.97\%}_{\text{intertemporal, } \mathbb{E}[k] \text{ moves}} + \underbrace{\iota \approx 0}_{\text{interaction}} = \underbrace{\Delta Y = 2.10\%}_{\text{total, of frictionless output}}. \tag{46}$$

The sensitivity of both legs to the returns-to-scale parameter γ is in Appendix OA.3 (Tables 10 and 11).

These output shortfalls are not welfare costs, because the map from output to welfare is nonlinear. The consumption-equivalent cost, the scaling of the composite $\tilde{c}$ in every state that gives the frictional household the frictionless economy's ergodic expected utility, is 1.4 percent, exact under log utility as $\exp(\mathbb{E}\log\tilde{c}^{\mathrm{fl}} - \mathbb{E}\log\tilde{c}) - 1$ on the two ergodic panels at the same σ_z. It is below the 2.09 percent output loss because the rationed hours are not worked. Lower capital also lowers the maintenance bill, so the 0.97% intertemporal output shortfall nets to a smaller consumption loss.[11]

[9] At fixed price gaps (x_r, x_w), both Y and Y^{CE} scale as k^{α}, so their ratio, and hence $\mathcal{L} = \log(Y^{\mathrm{CE}}/Y)$, is independent of k. The interaction is zero exactly, not numerically.

[10] $\sigma_z = 0.003091$ on the Kydland and Prescott convention. The frictionless arm of this decomposition carries the same σ_z, since the question is what the friction costs given the shocks. The frictionless column of Table 3 instead carries its own refit σ_z, since there the question is what each economy delivers at the same output volatility.

[11] With $\delta = 0.017715$ and $\Delta k = k^{\mathrm{CE}} - k^{*} = 0.30$, saved maintenance is $\delta\,\Delta k = 0.45\%$ of output, so the intertemporal consumption loss is 0.52% and the total consumption loss is 1.65% of output. This is a steady-state accounting comparison, not discounted risk-adjusted welfare.

OA.3 Grids, calibration, data, and robustness

This appendix gives the state space and its grids, the source and construction of every data series, the six-field record for each internally fitted parameter, and the robustness of the headline. It closes by placing the friction against the nearest quantitative literature.

OA.3.1 The state space and its grids

The state space is discretized. Capital sits on a log-spaced grid of 40 nodes bounded to $[0.45, 1.20]$ times its steady-state level, the two price gaps on symmetric grids of 21 nodes spanning ± 0.016 and ± 0.013, about four ergodic standard deviations, and the technology process on an eleven-state Markov chain fitted to the AR(1) by the method of Tauchen, its nodes spanning ± 3 unconditional standard deviations of $\log z$. The model constraint $k' \geq 0$ from (3) is enforced as the lower grid bound, about 0.45 times steady-state capital, which the ergodic distribution never approaches, so it does not bind in equilibrium.

The Tauchen chain enters only the solution step, where the conditional expectation in the Euler equation is taken as an eleven-node sum. The simulation instead draws continuous standard-normal innovations, $\log z_t = \rho_z \log z_{t-1} + \sigma_z \varepsilon_t$ with $\varepsilon_t \sim N(0,1)$, and reads the saving policy by interpolation in $\log z$ between the Tauchen nodes. This seam between a discretized solve and a continuous simulation is harmless, because the policy is interpolated continuously in $\log z$, the Tauchen nodes never restrict the simulated path, and the continuous draws leave the eleven-node support only about a quarter of a percent of the time, in the tails, where they are clipped to the grid edge before the policy is read.

OA.3.2 The calibration targets

Two parameters are fitted to a moment rather than matched by construction (γ and σ_z) or assigned from external micro evidence (ϕ_w). They are the rental persistence ϕ_r and the discount rate ρ. Table 6 records for each the data formula and value, the model formula and value, the sample, the standard error and the identifying elasticity.

Table 6: Target records for the two internally fitted parameters, ϕ_r and ρ.

field	ϕ_r (rental persistence)	ρ (discount rate)
NAME	binding-constraint ratio, labor-limited to equipment-limited plants	ergodic capital–output ratio ($\equiv$ gross investment share I/Y)
DATA FORMULA	US Census QPC (form MQ-C1, Sec. B) box 54 "insufficient local labor" share over box 56 "equipment limitations" share, All Manufacturing plant shares (marginal, boxes not mutually exclusive)	$E[K]/E[Y]$ annual, equivalently $E[I]/E[Y] = \delta \cdot K/Y$. US postwar
DATA VALUE	1.54 (moving-block bootstrap SE 0.084), 2013Q1–2019Q4, n=28, COVID excluded	$K/Y \in [2.4, 3.3]$ (mid 2.85), between the private-capital 2.35 of Khan and Thomas (2008) and the broad 3.32 of Cooley and Prescott (1995), a gross investment share $\delta K/Y = 0.202$ at the model's depreciation rate
MODEL FORMULA	$(\text{reg}_N + \text{reg}_{KN})/(\text{reg}_K + \text{reg}_{KN})$ under the stationary distribution, the ergodic probability labor binds over the ergodic probability capital binds	ergodic $E[k]/E[Y]$, deterministic $s_{ss} = \delta\alpha/(\rho + \delta)$, $K/Y = \alpha/[4(\rho + \delta)]$
MODEL VALUE	1.55 (a miss of about 0.09 SE), from ergodic regime shares $(0.481, 0.050, 0.217, 0.252)$	ergodic $K/Y = 2.850$. Deterministic $s_{ss} = 0.202$, $K/Y = 2.858$, at $\rho = 0.0106$/qtr (4.32%/yr)
ELASTICITY	$e[\mathcal{R}, \text{rental speed}] = +0.63$ (centered difference in $\log \lambda_r$ about the baseline calibration)	$e[K/Y, \rho] = -0.37$

NOTES: Each field is filled, and the last row is the elasticity of the target on the fitting parameter that identifies it.

The ratio target is a sample mean matched to an ergodic one, and three features of the data qualify it. First, it trends. Across 2013Q1–2019Q4 the ratio rises as the expansion tightened the labor market faster than the capital side. A linear fit gives a slope of +0.028 per quarter ($t = 6.60$, $R^2 = 0.63$), the annual mean climbing from about 1.08 in 2013 to 1.82 in 2019. The model ratio is a stationary, trendless ergodic object, so the period mean 1.54 is matched and the trend read as labor-market tightening outside the model. Second, the mapping carries a behavioral assumption beyond the marginal-probability arithmetic, that plants weigh a labor and a capital constraint symmetrically when reporting. If labor scarcity is the more salient reason to report as the market tightens, the ratio is biased up and the fitted rental made spuriously fast. Third, the sample excludes COVID, because the 2020–21 collapse in capacity utilization was a supply-chain and demand event the factor-rationing reading does not cover.

The rental- and wage-speed identification shares the spirit of the micro-moment identification the benchmark real-friction papers supply and is in one respect stronger. Dupraz et al. (2025) document the countercyclical incidence of downward nominal wage rigidity in micro

data rather than assume it, and Khan and Thomas (2013) pin capital irreversibility jointly to establishment-level investment moments. Here the wage speed rests on the same class of micro evidence, administrative payroll and SIPP wage durations, and the rental speed on a survey moment, the plant-level incidence of a binding labor versus capital constraint.

What the ratio fixes is an order-of-magnitude speed contrast rather than a point. The both-in-shortage cell reg_{KN} enters both marginals, so it discriminates nothing and only dilutes, compressing the pure one-sided ratio of 4.3 (the regime shares of Subsection 4.2, 0.217/0.050) into the marginal ratio $\mathcal{R}$ and destroying about 83% of the excess over one. What survives, and what moves with the speeds, is the excess of the one-sided mix above its geometric floor of about two, the floor itself fixed production and supply geometry, not the rental speed. The elasticity of $\ln \mathcal{R}$ is $+0.63$ to the rental speed and -0.35 to the wage speed (log differences about the baseline calibration with σ_z fixed), of opposite sign, so $\mathcal{R}$ tracks the contrast between the two markets' speeds net of the floor.

OA.3.3 Data sources

All targets are quarterly US series, logged with rates in levels and HP(1600)-filtered, each series' standard deviation, ratio to output, correlation with output and first autocorrelation recomputed from the source series. The output-volatility target that pins σ_z keeps the full window through 2020–21, since refitting it to 1947–2019 moves it only to $\text{std}(\log Y) = 0.016$. The binding-constraint target runs 2013Q1–2019Q4, with COVID excluded.

Series	std	rel-Y	corr-Y	AC1	source / construction
Output Y	0.0163	1.00	+1.00	0.79	real GDP, BEA
Consumption	0.0138	0.85	+0.79	0.72	real PCE, BEA
Investment	0.0707	4.34	+0.85	0.79	real gross private domestic investment, BEA
Capital	0.0038	0.23	+0.10	0.97	perpetual inventory, $\delta_{\text{ann}} = 0.069$
Hours	0.0209	1.28	+0.87	0.82	nonfarm business hours, BLS
Labor share	0.0108	0.67	−0.26	0.61	compensation/output, BLS
Real wage	0.0116	0.71	−0.03	0.72	real hourly compensation, BLS
TFP	0.0085	0.52	+0.53	0.73	Solow residual, $\alpha = 0.36$
Employment rate	0.96pp	—	+0.85	0.75	CPS, BLS

NOTES: Two constructed observables are flagged. *Capital* is a perpetual-inventory stock built from BEA real gross private domestic investment with the annual depreciation $\delta_{\text{ann}} = 0.069$, the model's own value. BEA differencing gives a negative implied δ because the investment series is total private while the published stock is nonresidential only, and the level of $\text{std}(\log K)$ is cosmetic since HP filtering removes it.

TFP is the Solow residual $\log Y - 0.36 \log K - 0.64 \log N$ with $\alpha = 0.36$ assigned, and its standard deviation is insensitive to α. FRED mnemonics for every series are in Table 7.

Table 7 lists the series behind each scorecard row, with its source, sample and transformation.

Table 7: Data provenance for the scorecard, Table 3, and the additional untargeted moments, with each series' source, sample, and transformation.

scorecard rows it feeds	source	sample	series and transformation
std($\log Y$), skew($\log Y$), skew($\Delta \log Y$), $\mathrm{AC}_1(Y)$, cond. sd & half-life ratio of Y	BEA/FRED[a]	1947Q1–2026Q1 | 1947–2019	real GDP, `GDPC1`. Log, HP(1600)
std(C)/std(Y), corr(C, Y), skew(C), cond. sd & half-life ratio of C	BEA/FRED[a]	1947Q1–2026Q1 | 1947–2019	real personal consumption expenditure, `PCECC96`. Log, HP(1600)
$\mathrm{AC}_1(C)$ (persistence only)	GRR[c]	1948Q1–2025Q3	real per-capita PCE on nondurables and services less gross housing, Gomme–Ravikumar–Rupert `real_pc_cm` (`gomme.csv`). Log, HP(1600)
std(I/Y), $\mathrm{AC}_1(I/Y)$, std(I)/std(Y), $\mathrm{AC}_1(I)$, corr(I, Y), skew(I), cond. sd & half-life ratio of I and of $s{=}I/Y$	BEA/FRED[a]	1947Q1–2026Q1 | 1947–2019	total real gross private domestic investment, `GPDIC1`, for the headline rows and the ratio $s{=}I/Y$. The *fixed*, *nonres.*, and *durables* investment variant rows use `A007RA3Q086SBEA` (fixed), `PNFIC1` (nonresidential, sample 2007Q1–) and `DDURRA3Q086SBEA` (consumer durables). Log for I, level for the ratio, HP(1600)
std(u), corr(u, Y), $\mathrm{AC}_1(u)$, skew(u), cond. sd of u	BLS/FRED[b]	1947Q1–2026Q1 | 1947–2019	civilian unemployment rate, `UNRATE` (quarterly average). Level in percentage points, HP(1600)
AC_1(LS), corr(LS, Y) at $k = 0$ and at the trough, std(LS)/std(Y)	BLS[b]	1947Q1–2026Q1 | 1947–2019	nonfarm business labor share wN/Y (compensation over output). Log, HP(1600)
corr(w, Y), std(w)/std(Y), $\mathrm{AC}_1(w)$	BLS[b]	1947Q1–2026Q1 | 1947–2019	real hourly compensation, nonfarm business. Log, HP(1600)
std(R)/std(Y), mean R, std(R), $\mathrm{AC}_1(R)$	GRR[c]	1947Q2–2024Q1, $n = 308$	return to capital, Gomme–Ravikumar–Rupert basis (`RET`). Annual percentage points (levels) and its cyclical component, HP(1600)
std($\log N^s$)/std($\log N$) and cond. sd of N and N^s, and the vacancy ratio of Section 5	BLS/FRED[b]	1948Q1–2026Q3 | 1948–2019	civilian labor force `CLF16OV`, household-survey employment `CE16OV`, and job openings `JTSJOL` (2000Q4–), quarterly averages of monthly levels. Logs, HP(1600), common sample of each pair
std($\log k$)/std($\log K$) and cond. sd of $\varkappa$ and k	Fed G.17, BEA[e]	1967Q1–2026Q1 | 1967–2019	total-industry capacity utilization `TCU`, and the capital stock by perpetual inventory from real private investment `GPDIC1`, benchmarked to the BEA net stock. Logs, HP(1600), common sample
skew($\varkappa$), corr($\varkappa, Y$), utilization	Fed G.17[e]	1948Q1–2024Q4 | 1948Q1–2019Q4	manufacturing capacity utilization, `CUMFNS` (total-industry `TCU`, 1967Q1–, is the cross-check reported in Subsection 6.4). Level, HP(1600)
Calibration target (Section 4), not an untargeted scorecard row			
binding-constraint ratio (fixes ϕ_r)	Census QPC[d]	2013Q1–2019Q4, $n = 28$	Quarterly Survey of Plant Capacity Utilization, reported reasons for operating below capacity, the share of plants marking insufficient local labor (box 54) over the share marking equipment limitations (box 56)

NOTES: Source keys match Table 3. [a] BEA National Income and Product Accounts via FRED, [b] Bureau of Labor Statistics, [c] Gomme et al. (2011) return-to-capital and nondurables-and-services consumption series, [d] US Census Quarterly Survey of Plant Capacity Utilization, [e] Federal Reserve G.17. Every series is cut to its stated sample before the HP(1600) filter, and the COVID quarters are dropped from the 1947–2019 sample. The model column is the baseline stationary simulation (300×1000 ergodic quarters).

Table 8 collects further moments that are not calibration targets, in the same construction and columns as Table 3, so the scorecard of Section 4 can point to them without carrying

them. The investment and rental first-order autocorrelations there sit against a frictionless RBC that already lies within 0.07–0.09 of the data, so their large negative percentages are small-denominator artifacts rather than economic content and restate one miss, shortened output memory, several times over. The remaining rows are collapsed duplicates and the wage volatility.

Table 8: Additional moments, not calibration targets, in the same construction and columns as Table 3.

moment	frictionless economy	data	model	% of gap closed
Investment and rental persistence (RBC already near the data)				
$\mathrm{AC}_1(R)$[c]	0.71	0.80	0.53	−200%
$\mathrm{AC}_1(I)$[a]	0.71	0.78	0.54	−243%
$\mathrm{AC}_1(I)$, fixed[a]	0.71	0.88	0.54	−100%
$\mathrm{AC}_1(I)$, nonres.[a]	0.71	0.87	0.54	−106%
Sign-conditional half-life ratios (large bootstrap SE, weak identification)				
sign-cond. half-life ratio, I[a]	0.92	0.67 | 0.80	0.54	+152% | +317%
sign-cond. half-life ratio, C[a]	1	0.73 | 1.96	0.74	+96% | *sign*
sign-cond. half-life ratio, s[a]	0.95	0.90 | 1.02	0.54	+820% | *sign*
Wage volatility, collapsed duplicates, and investment variants				
$\mathrm{std}(w)/\mathrm{std}(Y)$[b]	0.50	0.72	0.09	−186%
$\mathrm{corr}(I,Y)$, fixed[a]	0.99	0.79	0.99	0%
$\mathrm{corr}(I,Y)$[a]	0.99	0.82	0.99	0%
mean R, ann. pp[c]	4.29	5.83	4.28	*level*
$\mathrm{std}(R)$, ann. pp[c]	0.19	0.77	0.18	−2%
$\mathrm{std}(I)/\mathrm{std}(Y)$, nonres.[a]	2.88	2.72	2.26	*no gap*
$\mathrm{std}(I)/\mathrm{std}(Y)$, durables[a]	2.88	2.90	2.26	*no gap*
$\mathrm{AC}_1(w)$[b]	0.71	0.73	0.95	*no gap*

NOTES: The last column follows Table 3, so a figure above 100% means the friction overshoots the data, reported uncapped, and the *no gap* and *level* rows carry no percentage. The sign-conditional half-life ratios carry moving-block bootstrap standard errors of 0.4–1.0 (weak identification), and their frictionless economy is the certified equilibrium's near-unit half-life ratio (I at 0.96, C and s within two simulation standard errors of one, shown as 1). *Sign* marks a sample whose data ratio crosses the benchmark. Provenance is in Tables 3 and 7.

A few scoring conventions govern Tables 3 and 8. The boom-bust, skewness and half-life-ratio rows are sign-conditioned on HP(1600) data, the sign taken on cyclical output for the boom-bust and skewness rows and on each series' own sign for the half-life ratio. For output the frictionless economy is symmetric by construction, skewness 0 and boom-bust ratio 1. For investment, the saving rate and consumption the skewness and boom-bust benchmarks are the small residual asymmetries of the certified frictionless economy, computed at its own refit σ_z, and the unemployment, labor-share, and utilization rows keep the degenerate benchmark $u \equiv 0$, $\mathrm{LS} \equiv \beta$ and $\varkappa \equiv 1$.

A row within one data standard error of the benchmark is marked *no gap*. The two utilization rows compare the model capital-employed fraction $\varkappa = K/k$ to survey capacity utilization (the capacity-utilization discussion in Subsection 6.4), an order-of-magnitude sign comparison, and the labor-share correlogram row is scored at the data trough lag, $k = +2$ on 1947Q1–2026Q1 and $k = +3$ on 1947–2019, where the model trough sits at $k = 0$.

OA.3.4 Robustness

OA.3.4.1 The rental-half-life comparative static A slower rental market leaves a longer-lived, more volatile return gap. On the full nonlinear model the ergodic standard deviation of the realized return R rises from 0.000586 to 0.000827 per quarter as the rental half-life lengthens from 0.15 to 2.00 years, with the baseline (0.31 yr) at 0.000637.

The comparative static works through the idle-capital term alone. By Equation (43) the change in mean capital between two rental speeds is the change in U over $r^{\mathrm{CE}}\eta$, up to the second-order premium and the curvature residual, and the return dispersion does not enter. Across the sweep at fixed σ_z the mean idle fraction of the stock rises from 0.95% at a half-life of 0.15 yr to 1.90% at 2.00 yr, so U roughly doubles, from about 2.7 to 5.5×10^{-4} per quarter, and mean capital falls by about 1.8%. The capital constraint binds with probability 0.16 to 0.44 across the sweep and the both-in-shortage share rises from 0.13 to 0.40. The precautionary intuition, under which a more volatile return would raise saving and lift mean capital, does not apply here, because under log utility the prudence term is second order in a return standard deviation of order 10^{-3}.

OA.3.4.2 What rides on the wage half-life The wage speed is placed inside an interval, not at a point, so the moments it most directly governs should be checked across that interval. Those moments are the three headline unemployment statistics, the volatility, mean loss and skewness of unemployment, since the wage speed sets how much of a productivity innovation lands on the rationed employment quantity rather than on the posted wage. Table 9 re-solves the full nonlinear model at both ends of the assigned interval and at its midpoint, recalibrating at each point the two internal parameters the wage speed disturbs, the shock size σ_z to $\mathrm{std}(\log Y) = 0.0163$ by the Kydland–Prescott rule and the rental speed to the binding-constraint ratio 1.54, with ρ held at its baseline value (K/Y stays at 2.85 throughout).

Table 9: The three headline unemployment moments across the assigned wage-half-life interval.

wage half-life	ϕ_w	rental half-life	σ_z	std(u) pp	mean loss %	skew(u)
0.75 yr	0.794	0.248 yr	0.00334	1.441	1.044	1.306
0.85 yr	0.815	0.281 yr	0.00320	1.464	1.082	1.282
0.95 yr (baseline)	0.833	0.309 yr	0.00309	1.478	1.114	1.260

NOTES: The full nonlinear model re-solved at the two ends of the assigned interval and its midpoint, with σ_z refit to std($\log Y$) = 0.0163 and the rental speed refit to the binding-constraint ratio at each point, ρ held. The three moments are the unemployment statistics the wage speed governs most directly, the family the lever moves rather than an exhaustive robustness set.

Over the whole interval std(u) moves only 2.5% (from 1.441 to 1.478 pp), the mean loss from 1.044 to 1.114%, and skew(u) from 1.306 to 1.260, while the refit rental speed keeps the ratio of the two half-lives at 3.0 throughout. The headline is a property of the rationing mechanism, not of where inside its assigned interval the wage speed is placed. One limit qualifies the exercise, the level overshoot. Faster wage adjustment lowers std(u), yet even the fastest speed in the interval (half-life 0.75 yr) leaves std(u) = 1.44 pp, still about 1.4 times the data target near 1.0 pp, so no admissible ϕ_w closes that gap.

OA.3.4.3 Higher returns to scale The headline also survives higher returns to scale. The interior factor-demand elasticity is $\propto 1/(1-\gamma)$, rising from 10 at $\gamma = 0.90$ through 20 at $\gamma = 0.95$ to 50 at $\gamma = 0.98$, so the amplification of a thinner profit share is real, and under the Kydland–Prescott convention σ_z absorbs it, refit at each node to hold std($\log Y$) = 0.0163 ($\sigma_z = 0.0031$ at $\gamma = 0.90$, $\sigma_z = 0.0016$ at $\gamma = 0.95$). Table 10 re-solves the model at $\gamma = 0.95$ against its own frictionless economy, where unemployment volatility is 1.54 pp and the mean loss 1.25%, both close to the baseline values. Across the solvable range $\gamma \in [0.90, 0.95]$ the untargeted moments move only mildly, the unemployment volatility from 1.478 to 1.540 pp and the mean loss from 1.113% to 1.254%, so σ_z and γ are close to orthogonal for those moments. The calibrated speeds admit no solution at $\gamma = 0.98$, where the same elasticity reaches 50, so that upper end is not reported.

Table 11 extends the single $\gamma = 0.95$ head-to-head to the full solvable range. The size of both legs is governed by γ, which sets the pure-profit share $1-\gamma$ and, through it, the both-in-surplus amplification $1/(1-\gamma)$ of Equation (33). Because that regime supplies 92% of the mean static loss, at fixed price gaps the static leg would scale one for one with $1/(1-\gamma)$. The static leg rises from 0.99% at $\gamma = 0.85$ to 1.25% at $\gamma = 0.95$, a ratio of 1.26 rather than the 3.0 by which $1/(1-\gamma)$ itself rises, the idle fraction moves with it from 1.07% to 1.28%, and the intertemporal leg Δ_{INT} from 0.77% to 1.24%, so the total output loss climbs from 1.77% to 2.49%. What moves besides the $1/(1-\gamma)$ scaling is the refitted shock size, which

Table 10: Robustness at $\gamma = 0.95$ (pure-profit share 5%), against a frictionless economy with the friction switched off at the same shock.

moment	frictionless	this model	data
std($\log Y$)	0.0030	0.0163	0.0163
std(u) (pp)	0.00	1.540	0.95
mean u (pp)	0.00	—	—
mean loss (%)	0.00	1.254	—

NOTES: σ_z is refit to 0.0016 so the frictional model holds std($\log Y$) = 0.0163 at the baseline calibration. The std(u) and mean-loss entries are those of the baseline calibration, and the mean-u level is not reported there. The frictionless column switches the friction off, so its unemployment and loss entries are zero by construction.

the Kydland–Prescott rule lowers as γ rises to hold output volatility fixed, so the loss ratio understates the raw $1/(1-\gamma)$ factor.

Table 11: The static leg, idle capital, and the intertemporal leg across the returns-to-scale parameter γ.

γ	$1-\gamma$ (%)	static loss (%)	idle (%)	U (10^{-4}/qtr)	$\Delta \log \bar{k}$ (%)	Δ_{INT} (%)	total (%)
0.850	15.00	0.99	1.07	3.03	−1.84	0.77	1.77
0.900	10.00	1.11	1.16	3.32	−2.14	0.97	2.09
0.925	7.50	1.18	1.21	3.47	−2.31	1.09	2.27
0.950	5.00	1.25	1.28	3.67	−2.51	1.24	2.49

NOTES: Settled narrow-grid node, σ_z refit to std($\log Y$) = 0.0163 by the Kydland and Prescott rule at each γ. The static leg is the mean misallocation loss in the level convention $1 - Y/Y^{\text{CE}}$ of Section 4, so the $\gamma = 0.90$ row (1.11%, total 2.09%) is the level twin of the log-unit decomposition of Equation (46) (1.13%, total 2.10%), and Δ_{INT} is the intertemporal leg.

OA.3.5 The friction against the literature

OA.3.5.1 The fundamental surplus Ljungqvist and Sargent (2017) show that across otherwise disparate search-and-matching models the elasticity of market tightness with respect to productivity factors as $\Upsilon\, y/(y-x)$, where the first term is bounded by the matching elasticity and $y-x$ is the *fundamental surplus*, the part of a job's output the invisible hand is free to move to the job-creation margin ($x = z$ under Nash bargaining, $x = \hat{w}$ under a sticky wage). High unemployment volatility requires the fundamental-surplus *fraction* $(y-x)/y$ to be small, and, in Ljungqvist and Sargent (2017)'s recomputation of the Hagedorn and Manovskii (2008) calibration, std(u) $\approx$ 1.4 pp obtains only when that fraction is driven to about 0.04.

My model has no vacancies and no job-creation margin, so the statistic does not transfer mechanically. Its role is played by the elasticity of the rationed employment quantity to

productivity. In the fixed-capital log-linear diagnostic (the reported std(u) = 1.48 pp is fully nonlinear) a flexible wage carries a fraction $a_w = \nu m$ of a productivity innovation, with $m = 1/(1+\nu-\beta)$, while a frozen wage carries none, so the amplification is

$$\mathcal{A} \;=\; \frac{1/(1-\beta)}{m} \;=\; 1+\frac{\nu}{1-\beta} \;=\; 3.36, \qquad \frac{1}{\mathcal{A}} \;=\; 1-a_w \;=\; 0.30,$$

at $(\nu,\beta) = (1, 0.576)$. The rationed margin is labor demand itself, a first-order ($\approx \beta$) share of output mapping one-for-one into unemployment, and on impact the sticky wage passes through none of the innovation, so the whole shock lands on employment, and the slow wage (half-life 0.95 year, assigned interval 0.75–0.95 year) is what freezes it. In the terms of Ljungqvist and Sargent (2017) I substitute a slow adjustment speed for a small fundamental surplus.

OA.3.5.2 Idle capital and measured utilization In the model K/k is an extensive margin, the rationed firm renting a fraction of the installed stock and leaving the rest idle whenever the sticky rental exceeds capital's marginal product, with no intensity margin. Measured capacity utilization is a different construct, the variable intensity of use of installed capital of Greenwood et al. (1988), the margin this model omits, so the model idle-capital shortfall $(k-K)/k$ and one-minus-measured-utilization are not the same quantity.

What the two share is a signature, each a fraction of potential production actually realized and so left-skewed and procyclical, and that shared sign is all I test. On the same US sample as the calibration (manufacturing capacity utilization, 1948–2024, HP(1600), $N = 308$) cyclical capacity utilization is left-skewed, skew $= -0.80$, and procyclical, corr $= +0.89$ with cyclical output.[12]

OA.4 The firm block, its candidate set and uniqueness

The maximizer is unique. Taking productivity z, the predetermined capital bound k, the labor supply $N^s(w)$, and the posted prices r and w as given, the firm chooses (K, N) to solve the box-constrained profit maximization (8),

$$\max_{K,N}\; f(K,N) = zK^{\alpha}N^{\beta} - rK - wN \qquad \text{s.t.} \qquad 0 \le K \le k,\;\; 0 \le N \le N^s(w),$$

[12] FRED series `CUMFNS` and `TCU` (Federal Reserve G.17) and `GDPC1` (BEA), quarterly, HP(1600). The total-industry series gives -0.97 and $+0.87$ on 1967–2024. Greenwood et al. (1988) observe that constructed utilization series "convey ... similar information as a detrended ... production series," so this procyclicality is close to mechanical, read as consistent with the mechanism's qualitative signature, not as a measurement of the idle-capital object.

over the box $\mathcal{B} = [0, k] \times [0, N^s(w)]$. On the open positive orthant the Hessian of $zK^\alpha N^\beta$ is

$$z\,K^\alpha N^\beta \begin{pmatrix} \alpha(\alpha-1)/K^2 & \alpha\beta/(KN) \\ \alpha\beta/(KN) & \beta(\beta-1)/N^2 \end{pmatrix},$$

with determinant $z^2 K^{2\alpha-2} N^{2\beta-2}\,\alpha\beta\,(1-\alpha-\beta) > 0$ and negative diagonal, hence negative definite exactly because $\gamma = \alpha + \beta < 1$, and the linear term $-rK - wN$ leaves the Hessian unchanged, so f is strictly concave. A strictly concave function on the compact convex set $\mathcal{B}$ attains its maximum at a single point, and the KKT conditions are necessary and sufficient there. The maximizer is therefore the highest-f element of the KKT candidate set below. The lower edges $K = 0$ and $N = 0$ need no separate check, since the marginal product of each factor diverges as that factor vanishes, so any zero-factor point is dominated by the strictly positive profit at an interior or one-sided optimum.

There are four candidates. Writing $\mu_K, \mu_N \geq 0$ for the multipliers on $K \leq k$ and $N \leq N^s$, the stationarity conditions are $\alpha z K^{\alpha-1} N^\beta = r + \mu_K$ and $\beta z K^\alpha N^{\beta-1} = w + \mu_N$, and complementary slackness selects one of four cases. In the *interior* case, with $\mu_K = \mu_N = 0$, the firm sits at the unconstrained demands (K^d, N^d) that solve the two first-order conditions,

$$K^d = \left[z\,\alpha^{1-\beta}\beta^\beta\, r^{-(1-\beta)} w^{-\beta}\right]^{1/(1-\gamma)}, \qquad N^d = \left[z\,\alpha^\alpha \beta^{1-\alpha}\, r^{-\alpha} w^{-(1-\alpha)}\right]^{1/(1-\gamma)},$$

admissible when $K^d \leq k$ and $N^d \leq N^s$. In the *capital-short* case, with $K = k$ and $\mu_N = 0$, labor stays on its own first-order condition, $N = (\beta z k^\alpha / w)^{1/(1-\beta)}$ from (11), admissible when that $N \leq N^s$, and the shortfall $N^s - N > 0$ is involuntary unemployment. In the *labor-short* case, with $N = N^s$ and $\mu_K = 0$, capital stays on its own first-order condition, $K = (\alpha z (N^s)^\beta / r)^{1/(1-\alpha)}$ from (11), admissible when that $K \leq k$, and the shortfall $k - K > 0$ is idle capital. In the *both-bind* case both constraints hold with equality at the corner $(K, N) = (k, N^s)$.

The firm block evaluates f at every admissible candidate and returns the maximizer. The one point that a market-by-market rule would miss is that the capital-short and labor-short cases each leave the other factor strictly interior. The realized allocation is therefore not $\big(\min(K^d, k), \min(N^d, N^s)\big)$ applied separately to the two unconstrained schedules. It is the joint KKT solution, in which the unrationed factor is re-optimized at (11) against the binding one.[13]

[13] The realized allocation is continuous across the switch line where the two one-sided maps meet, at $x_r = (1 + 1/\nu) x_w$, since the unrationed factor is re-optimized against the binding one on both sides and the common value on that line is the interior optimum.

OA.4.1 The household's recursive problem has a unique solution

The atomistic household takes as given the stationary law of the aggregate state $s = (z, x_r, x_w)$, the posted-price rule (17), and the firm's rationed demands, and chooses only its own next-period capital k'. Writing $A(k, s) = (1 - \delta)k + Y(k, s)$ for the resources the budget (3) leaves once the profit rebate exhausts output, its problem on the compact grid is

$$V(k, s) = \max_{0 \le k' \le A(k,s)} \log\Big(A(k, s) - k' - \chi \tfrac{N^{1+\nu}}{1+\nu}\Big) + (1 + \rho)^{-1} \, \mathbb{E}\big[V(k', s') \mid s\big],$$

with the admissibility floor $\tilde{c} \ge \underline{\tilde{c}} > 0$ imposed, which bounds the log return above and below and removes its unboundedness at zero consumption. The Bellman operator T then satisfies the three conditions that make it a contraction. It maps bounded continuous functions to bounded continuous functions, since the return is continuous and bounded on the compact box. It is monotone, since a larger continuation value raises the maximand pointwise. And it discounts, since $T(V + a) = TV + (1 + \rho)^{-1}a$ for any constant a with $(1 + \rho)^{-1} < 1$. Its unique fixed point is the value function V.

The policy is unique as well. The resource map $A(k, s)$ is concave in k, flat where capital is in surplus and, where capital binds, equal to $z\, k^{\alpha/(1-\beta)}$ times a constant with $\alpha/(1-\beta) < 1$ because $\gamma < 1$, so its graph is convex and T carries concave functions to concave ones, making V concave in k. For a fixed state, consumption is linear in k', so the log return is strictly concave in k', and with the concave continuation term the objective is strictly concave in k' and its maximizer is a single point. The realized return $R' = 1-\delta+r'\varkappa'$ is kinked at the stock where next quarter's rationing begins, which removes differentiability but not concavity, so the maximizer stays unique.

This establishes a unique solution to the household's problem for a given aggregate law of motion. It does not establish that the recursive sticky-relative-price equilibrium is unique, since the map from an aggregate saving law into the household's best response is not a contraction in general, and the route by which frictionless models settle the question, supporting the allocation as the solution of a strictly concave planner's problem, is unavailable here because the allocation is not efficient.

OA.5 Solution and simulation

OA.5.1 Solving the model

The model is solved in discrete time, one quarter to the period, by time iteration on the optimizing household's saving policy, a function $k'(k, \log z, x_r, x_w)$ of the predetermined cap-

ital stock k, log technology $\log z$, and the two log price gaps $x_r = \log(r/r^{\mathrm{CE}}(z,k))$ and $x_w = \log(w/w^{\mathrm{CE}}(z,k))$ that carry the sticky rental and wage. Carrying the sticky prices as gaps to their competitive benchmarks makes the frictionless limit exact, since $\phi_r, \phi_w \to 0$ sends both gaps to zero.

Each iteration holds the continuation saving policy fixed and, at every grid node, solves the household intertemporal Euler equation (6) for the current saving choice k'. The argument of marginal utility is the GHH composite $\tilde{c} = c - \chi N^{1+\nu}/(1+\nu)$, and the frictional realized gross return $R' = 1-\delta+r'\varkappa'$ is built from the sticky rental r' and the rented fraction $\varkappa' = \min\{K'/k', 1\}$. The inner solve for k' is a bisection, not a Newton step, because the realized return has a corner in k' where the capital ceiling begins to bind, which a derivative-based root finder cannot cross reliably. The iteration is warm-started from the frictionless decreasing-returns policy and stops when the policy no longer moves, at $\sup|\Delta k'| < 10^{-8}$. Started instead from the no-adjustment policy $k' = k$, or from a policy that saves 10 percent less than the warm start at every node, the iteration returns the same policy to within 1.4×10^{-6}, about 10^{-7} of the steady-state stock.

Between the nodes the saving policy is read by multilinear interpolation on the four-dimensional grid, and the conditional expectation in the Euler equation is the eleven-node sum over the Tauchen chain. The frictionless economy is solved on its own two-dimensional state $(k, \log z)$ by the endogenous grid method, on 1000 capital nodes and a fifteen-node Tauchen chain of the same span, and is simulated with the same draws, sample length and filter as the frictional economy. One solve of the frictional economy takes about fifteen minutes on eight processes, the grid nodes being independent within an iteration. The replication package holds the code, the data and the solved policies, and regenerates every figure, table, and number in the paper.

OA.5.2 Simulating the model

The converged policy is simulated. Each quarter the technology innovation is realized, the two prices adjust part of the way toward their contemporaneous competitive values by the mechanical rule of Subsection 3.3, applied as a fixed law of motion,

$$\log p_t = \phi_p \log p_{t-1} + (1-\phi_p) \log p^{\mathrm{CE}}(z_t, k_t), \qquad \phi_p \in (0,1), \quad p \in \{r, w\},$$

the rationed firm block of Appendix OA.4 maximizes profit inside its box at the posted prices, trading on the short side of each factor market, and returns (K, N, Y), and the household then splits realized resources into consumption and next-period capital by (3) under its optimizing saving policy. Reported moments are computed over 300 simulated paths of

1000 retained quarters ($T = 1300$ with a 300-quarter burn-in), each series HP(1600)-filtered exactly as the data are.

The competitive targets $(r^{\mathrm{CE}}, w^{\mathrm{CE}}, N^{\mathrm{CE}})$ have the closed forms of Subsection 3.3, so no inner auctioneer loop is needed. The isolated transitions of Figures 2, 4, 6 and 7 start the economy at its rest point, the fixed point of the recursion with the innovation set to zero under the same stochastic saving policy, apply one ± 1 percent innovation to $\log z$ at date 0, and let $\log z$ decay at ρ_z with no further innovations, the household on its policy throughout.

The technology shock is calibrated last. Its standard deviation σ_z is refit on the full frictional model so that the simulated std$(\log Y)$ matches the data value 0.0163 at every parameter point, the Kydland and Prescott convention (Kydland and Prescott, 1982). The persistence $\rho_z = 0.95$ is a conservative reading, utilization-adjusted estimates running higher, Basu et al. (2006) treating technology as a random walk, King and Rebelo (1999) reporting 0.99 once capacity utilization is netted out, and Fernald (2014) providing the quarterly utilization-adjusted series.

OA.5.3 Accuracy

On the interior grid nodes the household Euler equation (6) holds to 1.3×10^{-8}. Along the simulated ergodic path, which lives between the nodes, the unconditional Euler restriction $1 + \rho = \mathbb{E}[R'] + \mathrm{Cov}(R', 1/\tilde{c}')/\mathbb{E}[1/\tilde{c}']$ closes to 6.5×10^{-5} per quarter on the reported grid. That residual is an interpolation error, largest where the saving policy kinks at a regime switch, and it is the one quantity that sets the level of the ergodic mean capital stock, so the discount rate is fitted on this grid to $\rho = 0.0106$ per quarter to hold the ergodic capital–output ratio at 2.85.

The identification is grid-robust. Refining the price-gap and technology grids leaves the binding-constraint ratio that fixes the rental speed between 1.54 and 1.56, within a quarter of its survey standard error (0.084) of the data target 1.54.

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